Investigation of Dielectric and Piezoelectric Properties of KNN($K_{0.5}Na_{0.5}NbO_3$) with BCHfT($Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$) Composite Ceramics

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SUPERVISOR'S APPROVAL

This is to certify that the thesis entitled "Investigation of Dielectric and Piezoelectric Properties of KNN ($K_{0.5}Na_{0.5}NbO_3$) with BCHfT ($Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$) Composite Ceramics" submitted by Pritom Majumder student no. 2011011 has been carried out under my supervision in the Department of Materials and Metallurgical Engineering, Bangladesh University of Engineering and Technology.

The thesis has been prepared in partial fulfillment of the requirements for the degree of Material and Metallurgical Engineering. To the best of my knowledge, the work presented in this thesis is original and has been completed with sincerity, dedication, and academic integrity. The thesis is hereby approved for submission and evaluation.

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CANDIDATE'S DECLARATION

I hereby declare that the thesis entitled "Investigation of Dielectric and Piezoelectric Properties of KNN ($K_{0.5}Na_{0.5}NbO_3$) with BCHfT ($Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$) Composite Ceramics" submitted by me, Pritom Majumder, to the Department of Materials and Metallurgical Engineering, Bangladesh University of Engineering and Technology, has been carried out as an original research work under proper academic supervision.

I further declare that this thesis has not been submitted, either wholly or partially, to any other university, institution, or organization for the award of any degree, or academic qualification. The information, data, figures, tables, and references used in this thesis have been properly acknowledged wherever necessary.

I take full responsibility for the authenticity, originality, and academic integrity of the work presented in this thesis.

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Date: ___________________________

DEDICATION

This thesis is dedicated to my beloved parents, whose endless love, sacrifice, encouragement, and blessings have been the greatest source of strength throughout my academic journey.

I also dedicate this work to my respected teachers, supervisor, and all those who have guided, supported, and inspired me with their knowledge, patience, and valuable advice.

Finally, this thesis is dedicated to everyone who believed in me and motivated me to continue my research with sincerity, dedication, and perseverance.

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Abstract

Lead zirconate titanate Pb(Zr, Ti)O3 (PZT) is one of the well-known ceramics possessing excellent piezoelectric performance. However, due to the presence of harmful lead component, much research attention has been paid in finding the lead-free piezoelectric ceramic with adequate dielectric and piezoelectric behaviors. Potassium sodium niobate $K_{0.5}Na_{0.5}NbO_3$ (KNN) was chosen in this work as the main lead-free piezoelectric matrix with a composite modification in the form of $Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$ (BCHfT).Pure KNN, pure BCHfT, and KNN mixed with 1 wt% BCHfT and 1.5 wt% BCHfT were prepared by the conventional solid-state method. Sintering process was carried out at 1090°C for 10 hours. The prepared materials were characterized by using X-ray diffraction technique for phase investigation; scanning electron microscopy (SEM) for microstructure investigation; frequency and temperature dependent dielectric studies; impedance/conductivity measurement; and d33 measurement for evaluation of piezoelectric performance. Based on dielectric measurements, all ceramics showed decreasing trend in real permittivity with increasing frequency because of the effect of fast relaxation polarization on the frequency dependence of the dielectric constant. From among the compositions investigated in this study, KNN-1 wt% BCHfT gave optimum values of dielectric and piezoelectric constants with relatively higher value of permittivity, better dielectric loss values than others, and maximum unpoled d33 value of 15.5 pC/N. On the other hand, KNN-1.5 wt% BCHfT exhibited higher dielectric loss especially in lower frequencies owing to the higher influence of defects and interface polarization. For KNN-1 wt% BCHfT material, thermally activated conductivity was observed along with diffuse dielectric characteristics. In summary, the incorporation of the compositional modifier in the lead-free piezoelectric ceramic leads to tuning of both dielectric and piezoelectric behaviors. Based on results obtained in this work, KNN-1 wt% BCHfT shows more promise compared to other compositions and further optimization may be considered regarding the sintering procedure, density adjustment, defect reduction, quality of electrodes, and proper poling.

Keywords: Lead-free piezoelectric ceramics; KNN; BCHfT; KNN–BCHfT composite; dielectric properties; piezoelectric coefficient; solid-state reaction; dielectric loss; AC conductivity; d_{33}.

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Chapter: 1 - Introduction

1.1 Background

Dielectric materials are an important class of functional materials because they can store electrical energy through polarization under an applied electric field. They are widely used in capacitors, sensors, actuators, transducers, resonators, communication systems, memory devices, and energy-storage applications. In dielectric ceramics, the real part of permittivity, dielectric loss, AC conductivity, impedance response, and temperature-dependent dielectric behavior are important parameters for evaluating polarization ability, energy dissipation, defect contribution, and electrical stability. Therefore, dielectric characterization is essential for understanding the functional performance of electroceramic materials.

Piezoelectric ceramics are closely related to dielectric and ferroelectric materials because their electromechanical response is strongly influenced by polarization behavior, dielectric permittivity, dielectric loss, domain-wall motion, and phase stability. Piezoelectric materials can convert mechanical strain or stress into electrical voltage and can also produce mechanical deformation under an applied electric field. For this reason, they are used in sensors, actuators, ultrasonic transducers, vibration sensing, precision motion control, biomedical devices, and energy-harvesting systems.[1] However, efficient piezoelectric performance cannot be understood only from the piezoelectric coefficient; the dielectric response of the material is also very important because polarization, dielectric loss, leakage current, and conductivity directly affect the stability and efficiency of piezoelectric devices.

For many decades, lead zirconate titanate, $Pb(Zr,Ti)O_3$ or PZT, has dominated the piezoelectric ceramic market because of its excellent dielectric, piezoelectric, and electromechanical coupling properties. However, the high lead content of PZT raises serious environmental and health concerns during processing, application, recycling, and disposal. These concerns have motivated extensive research into environmentally friendly lead-free ceramic materials with balanced dielectric and piezoelectric properties.[2]

Among the various lead-free piezoelectric candidates, potassium sodium niobate, $K_{0.5}Na_{0.5}NbO_3$ (KNN), has attracted considerable attention as a promising PZT alternative because of its perovskite structure, environmental compatibility, and useful ferroelectric-piezoelectric properties. However, pure KNN generally shows relatively low piezoelectric response and its processing is challenging because alkali volatilization during sintering can cause poor densification, low density, and microstructural degradation, thereby limiting its piezoelectric performance.[2] Barium titanate-based ceramics are another important family of lead-free ferroelectric materials. In particular, $Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$, abbreviated as BCHfT, is a modified $BaTiO_3$-based ceramic where Ca and Hf substitutions help tune phase transitions and improve dielectric and piezoelectric behavior.[3]

Therefore, forming a KNN–BCHfT composite is a promising strategy to improve the structure, microstructure, densification, dielectric response, and piezoelectric performance of lead-free ceramics. In this study, KNN–BCHfT composites are investigated as environmentally friendly piezoelectric materials for advanced smart-device applications.

1.2 Research Motivation

The motivation for developing lead-free piezoelectric ceramics is driven by the need to reduce dependence on lead zirconate titanate, $Pb(Zr,Ti)O_3$ (PZT), which has long been the dominant piezoceramic for electromechanical devices such as actuators, sensors, and transducers because of its excellent functional properties. However, the increasing use of PZT leads to the release of lead-containing compounds into the environment during processing, machining, recycling, and waste disposal, creating environmental and health concerns. These concerns, together with hazardous-substance regulations such as WEEE and RoHS, have strongly accelerated the search for high-performance lead-free piezoceramics as safer alternatives for future smart electronic and electromechanical applications.[4]

KNN is a strong candidate for this purpose, but pure KNN does not always show sufficient piezoelectric performance for practical applications. Its major limitations include processing difficulty, alkali-metal volatilization, poor densification, possible secondary-phase formation, and sensitivity of phase transitions. These factors can reduce density, interrupt grain growth, increase defects, and restrict domain-wall movement. Since domain-wall mobility and stable polarization are essential for high piezoelectric output, improving the structure and microstructure of KNN is necessary.[2] The motivation of this thesis is therefore to enhance KNN by introducing BCHfT as a composite component rather than relying only on pure KNN.

BCHfT is chosen because it is also a lead-free perovskite ferroelectric and because its Ca^{2+} and Hf^{4+} substitutions can effectively tune the rhombohedral-orthorhombic and orthorhombic-tetragonal phase transitions near room temperature, leading to improved dielectric and piezoelectric responses. In the KNN–BCHfT composite, BCHfT is expected to influence the phase boundary, microstructure, densification, and dielectric stability of the KNN-based system.[5]

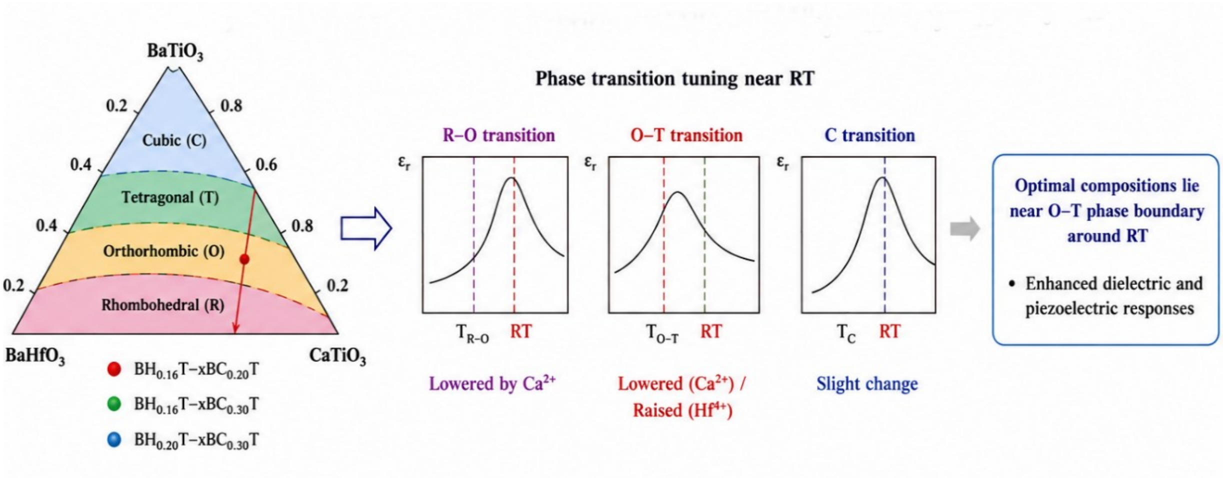


Figure 1.2.1: Tuning of Phase Transition in BCHfT. [5]

Another motivation is the need to improve stress-to-electric signal conversion. For sensing and energy-harvesting applications, a material must generate a strong and stable electrical signal under mechanical stress. This requires not only a high piezoelectric coefficient but also good density, stable polarization, controlled grain morphology, and suitable dielectric response.[6] The KNN–BCHfT composite may provide a better balance of these properties by combining the advantages of a KNN matrix with the phase-tuning and polarization-enhancing ability of BCHfT. This makes the composite system a meaningful route for developing environmentally friendly piezoelectric ceramics for future sensors, transducers, and low-power energy-harvesting devices.

The research is also motivated by a clear experimental gap. Many studies have focused on KNN-based doping, KNN-based texturing, and $BaTiO_3$-derived phase-boundary systems, but the specific composite approach using KNN with BCHfT still requires systematic investigation. The present thesis therefore aims to study how BCHfT addition affects the phase structure, microstructure, densification, ferroelectric behavior, dielectric response, and piezoelectric performance of KNN-based ceramics. The planned characterization methods: XRD for phase and crystal structure, FE-SEM/EDS for microstructure and elemental analysis, d_{33} meter and impedance analyzer for piezoelectric properties, and dielectric testing with frequency and temperature are directly aligned with this research objective.

The major goal of this thesis is to develop and investigate the lead-free KNN-BCHfT composite ceramic material with advantages over pure KNN in order to achieve improved dielectric and piezoelectric properties. The understanding of relationships between the processes of composite synthesis, phase development, densification, domain wall mobility, dielectric properties stability, and strain-to-potential conversion will be obtained with the intention of making a contribution to developing safer and more effective lead-free piezoelectric materials.

Chapter: 2 - Literature Review

2.1 Introduction

Functional materials that include piezoelectric materials form an important category of such materials because they possess the ability to convert mechanical energy into electrical energy and vice versa. The electromechanical effects make them suitable for a broad range of applications from sensors to energy harvesters. Lead zirconate titanate ($PbZr_{1-x}Ti_xO_3$) or simply PZT has been the most commonly used piezoelectric material in the form of ceramics for many decades due to its excellent dielectric and electromechanical properties. However, the presence of large amounts of lead oxide in the composition makes it environmentally unfriendly and poses significant health risks. Therefore, high performance lead free piezoelectric ceramics have attracted much attention.

Among different lead-free piezoelectric systems, potassium sodium niobate ($K_{0.5}Na_{0.5}NbO_3$ or KNN) has attracted considerable attention as a promising alternative to PZT. KNN possesses a perovskite crystal structure, good ferroelectric behavior, relatively high Curie temperature, and useful dielectric and piezoelectric properties. However, pure KNN has several limitations, including poor densification, narrow sintering temperature window, alkali volatilization during high-temperature sintering, abnormal grain growth, secondary phase formation, and unstable piezoelectric response. These problems can reduce density, disturb phase formation, increase defects, restrict domain-wall motion, and finally decrease the piezoelectric performance of KNN ceramics.

To overcome these limitations, different modification strategies such as doping, texturing, sintering-aid addition, phase-boundary engineering, and composite formation have been widely studied. In the present work, KNN is combined with $Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$, abbreviated as BCHfT. BCHfT is a $BaTiO_3$-based lead-free ferroelectric ceramic in which Ca^{2+} substitution at the A-site and Hf^{4+} substitution at the B-site can tune phase transitions, modify lattice distortion, improve dielectric stability, and enhance piezoelectric behavior. Therefore, the formation of a KNN–BCHfT composite is expected to improve densification, microstructure, phase coexistence, dielectric response, domain-wall mobility, and stress-to-electric signal conversion.

Dielectric investigation of KNN–BCHfT ceramics is important because phase-boundary engineering and dopant-induced phase transitions strongly influence polarization behavior and the dielectric–piezoelectric response of KNN-based piezoceramics. Moreover, KNN materials with high dielectric constant and low dielectric loss have been reported to be useful for capacitors and energy-storage devices, while their piezoelectric response makes them suitable for sensors, actuators, and transducers.[7]

This literature review discusses the background, historical development, dielectric fundamentals, polarization mechanisms, and structure–property relationships of lead-free piezoelectric ceramics. Special attention is given to KNN-based ceramics, BCHfT as a $BaTiO_3$-based modifier, and the scientific importance of KNN–BCHfT composite formation. The review also highlights the role of phase structure, microstructure, densification, dielectric behavior, and piezoelectric response in developing environmentally friendly lead-free ceramics for advanced smart-device applications.

2.2 History of Dielectricity and Piezoelectricity

The historical development of piezoelectric ceramics is closely related to the development of dielectric science, because both phenomena are based on the response of electric polarization to external stimuli. Dielectricity deals with the ability of an insulating material to become polarized under an applied electric field, whereas piezoelectricity describes the coupling between mechanical stress and electric polarization in non-centrosymmetric materials. Therefore, before discussing the history of piezoelectric ceramics, it is important to briefly discuss the historical development of dielectricity, since dielectric permittivity, dielectric loss, polarization, relaxation, and conductivity are fundamental parameters for understanding ferroelectric and piezoelectric ceramic materials.

Historically, the development of dielectric materials is closely connected with the evolution of capacitor technology. A capacitor was initially understood as a simple device consisting of an insulating medium placed between two conductors. Later, Faraday made major contributions to capacitor technology by introducing the concept of dielectric constant, showing that the insulating material between conductors plays an active role in determining capacitance. This understanding established dielectrics not merely as non-conducting materials, but as functional materials capable of storing electrical energy under an applied electric field. Therefore, the dielectric constant became an important parameter for evaluating the energy-storage ability of dielectric materials.[8]

Maxwell extended Faraday's experimental concepts into a mathematical theory of the electromagnetic field, in which electrical and magnetic phenomena were explained through actions occurring in the surrounding medium. In this theory, when an electromotive force acts on a dielectric material, it produces electric polarization and a displacement of electricity within the dielectric. Maxwell further explained that variations of this electric displacement behave as currents, linking dielectric polarization with electromagnetic field behavior. His electromagnetic theory also showed that disturbances in the field can propagate as transverse waves with a velocity close to that of light, thereby establishing the importance of dielectric media in electromagnetic wave propagation and field theory.[9]

After the macroscopic description of dielectric behavior was established, later theoretical studies explained dielectric permittivity from the microscopic response of matter. Brouder and Rossano showed that the macroscopic dielectric constant of a periodic medium can be calculated from the microscopic dielectric function by considering polarization and local-field effects. Their formulation also connects this microscopic–macroscopic relationship with the Clausius–Mossotti or Lorentz–Lorenz type relation. This indicates that dielectric response is not only a bulk property, but also depends on the way charges, atoms, ions, and dipoles respond locally to an applied electric field inside the material.[10]

A major advancement in dielectric science occurred when Peter Debye introduced the theory of dielectric relaxation. Debye showed that permanent dipoles require a finite time to align with an alternating electric field. Therefore, dielectric permittivity is not a constant value for all conditions; rather, it depends strongly on frequency and temperature. At low frequency, slow polarization mechanisms such as dipolar and space charge polarization can contribute significantly to the total dielectric response. However, at higher frequency, these slower mechanisms cannot follow the rapidly changing electric field, resulting in a decrease in dielectric constant. This concept is directly relevant to ceramic dielectrics, where electronic, ionic, dipolar, interfacial, and space-charge polarization mechanisms may contribute differently at different frequencies.[11]

The discovery of ferroelectric behavior established an important connection between dielectricity and piezoelectricity. In his study of Rochelle salt, Valasek showed that the polarization of the crystal exhibits an electric hysteresis loop similar to magnetic hysteresis in iron, indicating the presence of permanent polarization. He also reported that the dielectric displacement and the dielectric constant of Rochelle salt depend on the applied electric field. In addition, the piezoelectric response changed under different applied electric fields, showing that electrical and mechanical effects are closely related in this material. These observations provided an early experimental basis for understanding switchable polarization in ferroelectric

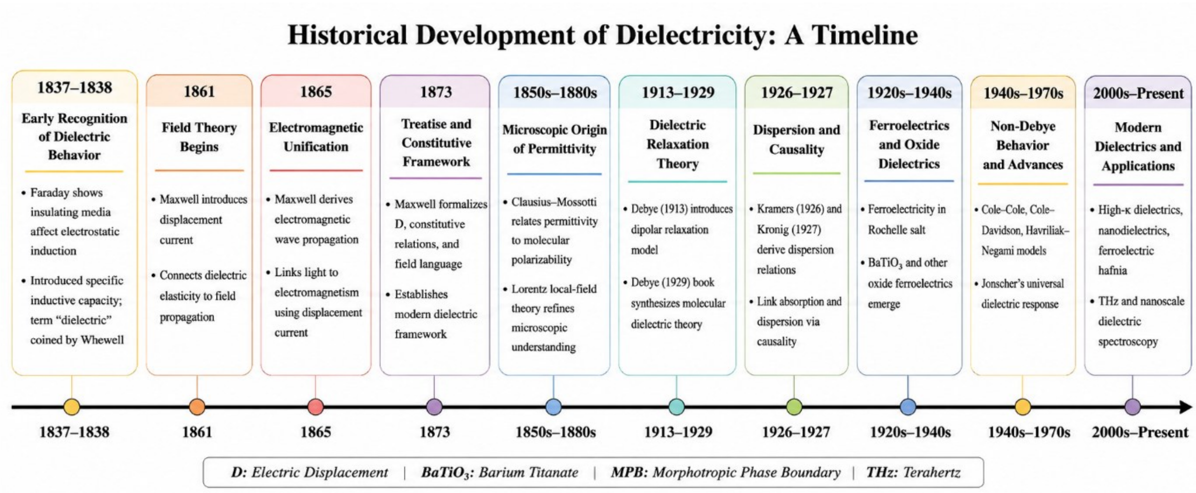


Figure 2.2.1: Historic timeline of Dielectricity.

materials, which later became important for explaining the dielectric and piezoelectric behavior of ferroelectric ceramics.[12]

The history of piezoelectric ceramics is closely related to the development of modern electroceramic materials and their applications in sensors, actuators, ultrasonic transducers, energy-harvesting devices, and smart electronic systems. Piezoelectricity is an electromechanical coupling phenomenon observed in non-centrosymmetric materials, where applied mechanical stress produces electric polarization or charge generation, while an applied electric field produces mechanical deformation through the inverse piezoelectric effect.[13] This coupling principle remains the scientific foundation for modern lead-free piezoelectric ceramic systems, including potassium sodium niobate modified with barium calcium titanate hafnate, commonly expressed as KNN–BCHfT.

Piezoelectricity was first discovered by Jacques Curie and Pierre Curie in 1880, when they demonstrated that mechanical compression of asymmetric crystals along their hemihedral axes generated electric polarity. In 1881, Lippmann theoretically predicted the converse piezoelectric effect, which was later confirmed experimentally by the Curie brothers.[14] The historical development of piezoelectric ceramics then progressed through the study of early ferroelectric crystals. In the 1920s, Joseph Valasek's investigation of Rochelle salt revealed dielectric hysteresis and switchable spontaneous polarization, introducing the concept of ferroelectric domain switching. This discovery was important because domain reorientation later became a key mechanism for explaining the high piezoelectric response of ceramic materials. However, early ferroelectric crystals such as Rochelle salt were fragile, water-soluble, and environmentally unstable, which limited their practical applications.[15]

A major advancement occurred in the 1940s with the discovery of barium titanate, $BaTiO_3$, which marked the beginning of the perovskite electroceramic era. $BaTiO_3$ became the first widely recognized oxide ferroelectric ceramic, offering better chemical stability and manufacturability than earlier single-crystal materials. Its perovskite structure enabled spontaneous polarization and ferroelectric switching. More importantly, the discovery that polycrystalline $BaTiO_3$ ceramics could be electrically poled showed that randomly oriented grains could be converted into a macroscopically piezoelectric material by aligning ferroelectric domains under a strong electric field. This concept became the foundation of modern piezoceramic processing.[16]

Although $BaTiO_3$ was historically significant, its relatively low Curie temperature and limited piezoelectric performance restricted its use in demanding applications. In the 1950s, lead zirconate titanate, $Pb(Zr,Ti)O_3$ or PZT, became the next major breakthrough. In 1954, Jaffe, Roth, and Marzullo reported that $PbZrO_3$–$PbTiO_3$ solid-solution ceramics showed superior piezoelectric properties, especially near the rhombohedral–tetragonal phase boundary, where electromechanical response increased sharply and the composition $Pb(Zr_{0.55}Ti_{0.45})O_3$ showed high coupling and temperature stability.[17]

Despite its excellent functional properties, PZT contains toxic lead, which raises environmental and health concerns during processing, use, and disposal. Environmental regulations such as RoHS, WEEE, and REACH accelerated the search for lead-free piezoelectric alternatives. Among different lead-free candidates, potassium sodium niobate, $K_{0.5}Na_{0.5}NbO_3$ or KNN, became one of the most promising systems because of its perovskite structure, relatively high Curie temperature, strong ferroelectric behavior, and environmental compatibility. In 1959, Egerton and Dillon investigated the dielectric and piezoelectric properties of potassium–sodium niobate ceramics and showed that compositions near the 50:50 K/Na ratio exhibited useful electromechanical coupling. However, KNN also showed limitations such as poor densification, high dielectric loss, alkali volatilization, and difficulty in microstructural control during sintering.[18]

For many years, KNN remained less attractive than PZT because of its processing difficulty and lower property stability. A major breakthrough was reported by Saito et al. in 2004, who developed alkaline niobate-based textured ceramics by combining morphotropic phase-boundary design with <001> oriented microstructural engineering. Their textured ceramic showed a peak d_{33} value of 416 pC/N and field-induced strain comparable to actuator-grade PZT, proving that properly engineered KNN-based ceramics could become promising environmentally friendly alternatives to lead-based piezoceramics.[19] Since then, KNN research has focused on phase-boundary engineering, dopant modification, densification improvement, and microstructural control.

In parallel, $BaTiO_3$-derived lead-free systems also developed significantly. Modified $BaTiO_3$ ceramics containing calcium, zirconium, tin, or hafnium attracted attention because they can tune phase-transition behavior and improve dielectric and piezoelectric responses. The BCHfT composition, $Ba_{0.85}Ca_{0.15}Ti_{0.9}Hf_{0.1}O_3$, belongs to this $BaTiO_3$-based family. Calcium substitution at the A-site and hafnium substitution at the B-site can modify lattice distortion, polarization behavior, dielectric response, and phase stability.[20] Therefore, combining BCHfT with KNN is a logical lead-free strategy, where KNN provides high Curie temperature and ferroelectric potential, while BCHfT can act as a phase-transition tuner, microstructural modifier, and dielectric-stability enhancer. Thus, KNN–BCHfT composite ceramics are scientifically important as a modern lead-free approach that combines phase-boundary engineering, densification improvement, dielectric modification, and environmentally sustainable piezoelectric material design.

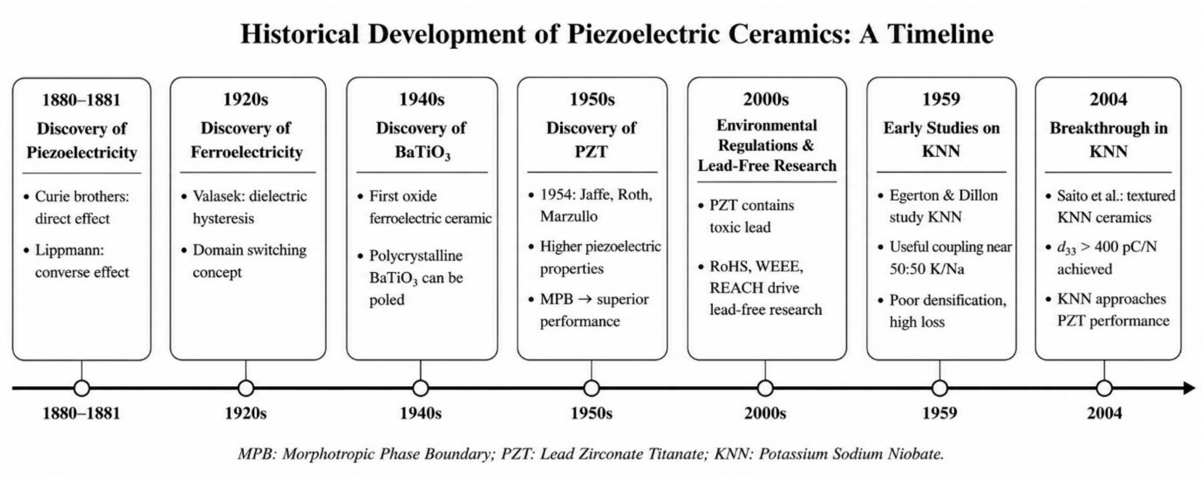


Figure 2.2.2: Historic timeline of piezoelectric ceramics.

2.3 Basic Theory of Piezoelectricity

Piezoelectricity appears in non-centrosymmetric crystalline materials. In the direct piezoelectric effect, mechanical stress produces electrical charge or polarization. In the converse piezoelectric effect, an applied electric field produces mechanical strain. These two effects are the foundation of piezoelectric sensors, actuators, transducers, and energy-harvesting devices.[13]

The linear constitutive equations of piezoelectricity can be expressed as:

$$D_i = d_{ij}T_j + \varepsilon_{ij}^T E_j \quad \textbf{(1)}$$
$$S_i = s_{ij}^E T_j + d_{ij}E_j \quad \textbf{(2)}$$

And its stress-charge form is:

$$T_j = C_E S_i - e^t E \quad \textbf{(3)}$$

$$D_i = eS + \varepsilon_s E \quad \textbf{(4)}$$

Where $D_i$is electric displacement, $S_i$is mechanical strain, $T_j$is mechanical stress, $E_j$is electric field, $d_{ij}$is the piezoelectric charge coefficient, s_E is the compliance tensor under constant electric field, c_E is the stiffness tensor under constant electric field, $\varepsilon_{ij}^T$is dielectric permittivity at constant stress, and $s_{ij}^E$is elastic compliance at constant electric field. For ceramic piezoelectric materials, the most commonly discussed coefficient is d_{33}, which represents the charge generated along the poling direction when mechanical stress is applied in the same direction. The superscript (t) indicates the transpose of the tensor.[21]

For energy harvesting and sensing applications, $d_{33}$is not the only important parameter. The piezoelectric voltage coefficient $g_{33}$is also important because it indicates the electric field or voltage generated per unit mechanical stress. It can be approximated by:

$$g_{33} = \frac{d_{33}}{\varepsilon_0 \varepsilon_r} \quad \textbf{(5)}$$

where $\varepsilon_0$is the permittivity of free space and $\varepsilon_r$is the relative dielectric permittivity. A common energy-harvesting figure of merit is:

$$FOM = d_{33} \times g_{33} = \frac{d_{33}^2}{\varepsilon_0 \varepsilon_r} \quad \textbf{(6)}$$

Thus, a good piezoelectric energy harvester requires not only a high d_{33}, but also controlled dielectric permittivity and low dielectric loss.[22] This is important for the KNN–BCHfT system because BCHfT may improve d_{33}, but excessive dielectric constant or dielectric loss could reduce voltage generation efficiency.

2.4 Basic Theory of Dielectricity

Dielectric materials act as polarizable insulating media in capacitors, where the application of an external electric field induces electric polarization and charge separation. During this charging process, electrical work is stored as electric energy within the dielectric, and the energy-storage capability is strongly related to the dielectric permittivity and the applied electric field.[23]

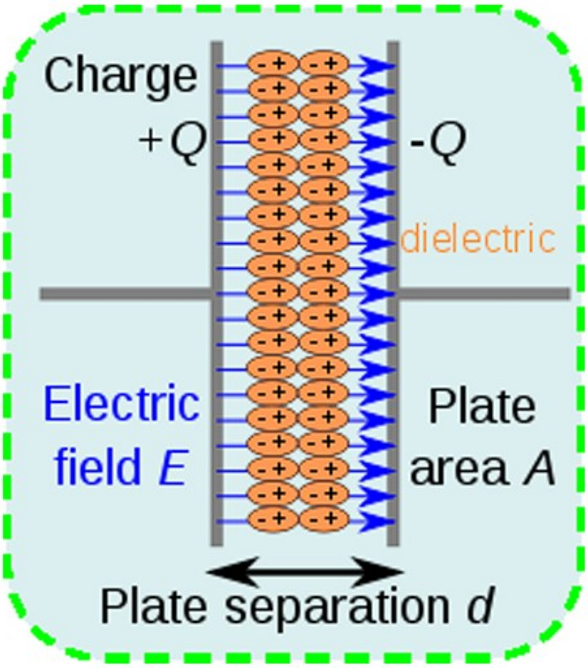


Figure 2.4.1: The diagram of charge separation in parallel-plate capacitor under the function of electric field.[23]

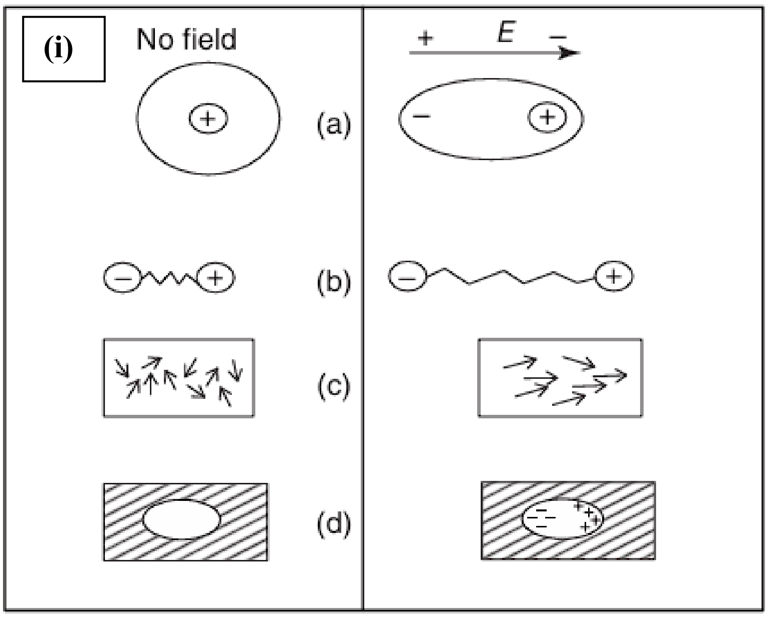
The terms "insulator" and "dielectric" are often used together, but they are not exactly the same. An insulator mainly refers to a material with very low electrical conductivity, whereas a dielectric material is generally poor conductors of electric current and are often regarded as electrical insulators; however, their functional importance lies in their ability to become polarized under an applied electric field. This polarization allows dielectric materials to store electrical energy, and their dielectric properties, particularly the dielectric constant, are important parameters for evaluating their energy-storage behavior.[24]

Dielectric materials are widely used in capacitors, sensors, transducers, resonators, actuators, memory devices, communication systems, and energy-storage devices. In dielectric capacitors, dielectric materials are placed between two conducting plates, and their energy-storage ability is governed by the permittivity of the dielectric medium. When an external voltage is applied, electric polarization occurs within the dielectric, leading to charge accumulation on the capacitor plates. During this charging process, electrical work is converted into stored electric energy in the dielectric, making dielectric polarization a key mechanism for capacitor-based energy-storage applications.[23]

Dielectric polarization occurs through different mechanisms:

1) **Electronic polarization** occurs when the electron cloud of an atom is displaced relative to the nucleus.

2) **Ionic polarization** occurs in ionic materials when positive and negative ions shift in opposite directions under an electric field.
3) **Dipolar/ Orientational polarization** occurs in materials with permanent dipole moments, where dipoles try to align with the applied field.
4) **Space charge polarization** occurs due to the accumulation of charges at interfaces, grain boundaries, defects, or electrode regions.



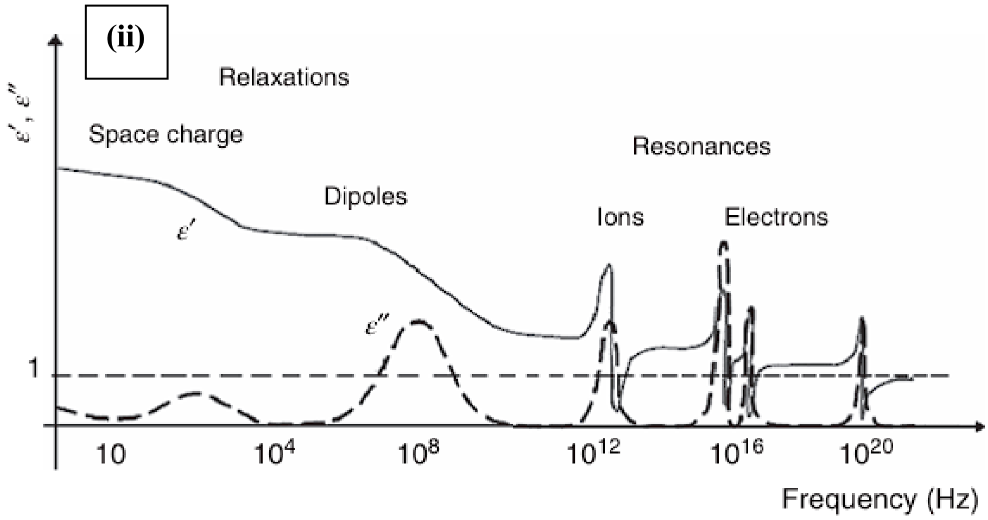


Figure 2.4.2: i) Polarization processes: (a) electronic polarization, (b) ionic polarization, (c) orientational polarization, and (d) space charge polarization. Ii) Frequency dependence of polarization dispersion.[25]

The total polarization of a dielectric material is the combined effect of these polarization mechanisms.(equation: 7)[25]

$$P_{total} = P_e + P_i + P_o + P_s \qquad \textbf{(7)}$$

where:

P_e= electronic polarization

P_i= ionic polarization

P_o= orientational polarization

P_s= space charge polarization

The dielectric response of a material is strongly influenced by both frequency and temperature because different polarization mechanisms possess different relaxation times. At lower frequencies, electronic, ionic, orientational, and space-charge polarization can contribute to the total polarization. However, with increasing frequency, the slower polarization mechanisms, particularly space-charge and orientational polarization, are unable to follow the rapidly alternating electric field, resulting in a decrease in dielectric constant. Temperature also affects the dielectric behavior, since thermal energy can randomize the alignment of permanent dipoles and promote ion diffusion associated with space-charge polarization.[25]

Dielectric loss is another important property of dielectric materials. It represents the energy dissipated as heat when the material is subjected to an alternating electric field. Low dielectric loss is desirable for capacitors, sensors, resonators, and piezoelectric devices because excessive loss reduces efficiency and may indicate leakage current, defects, poor densification, or conducting grain boundaries. Therefore, an ideal dielectric material should have high dielectric constant, low dielectric loss, good insulation resistance, high thermal stability, and stable frequency response.[26]

In ferroelectric and piezoelectric ceramics, dielectric properties are especially important because they are directly related to polarization switching, domain-wall motion, phase transition, and electromechanical performance.[27] Materials such as KNN and BCHfT are dielectric as well as ferroelectric/piezoelectric in nature. In the KNN–BCHfT composite system, dielectric analysis can help explain how BCHfT addition affects phase stability, densification, polarization behavior, dielectric loss, and piezoelectric performance. Therefore, the study of dielectrics provides an essential theoretical foundation for understanding the electrical and functional properties of lead-free KNN–BCHfT ceramics.

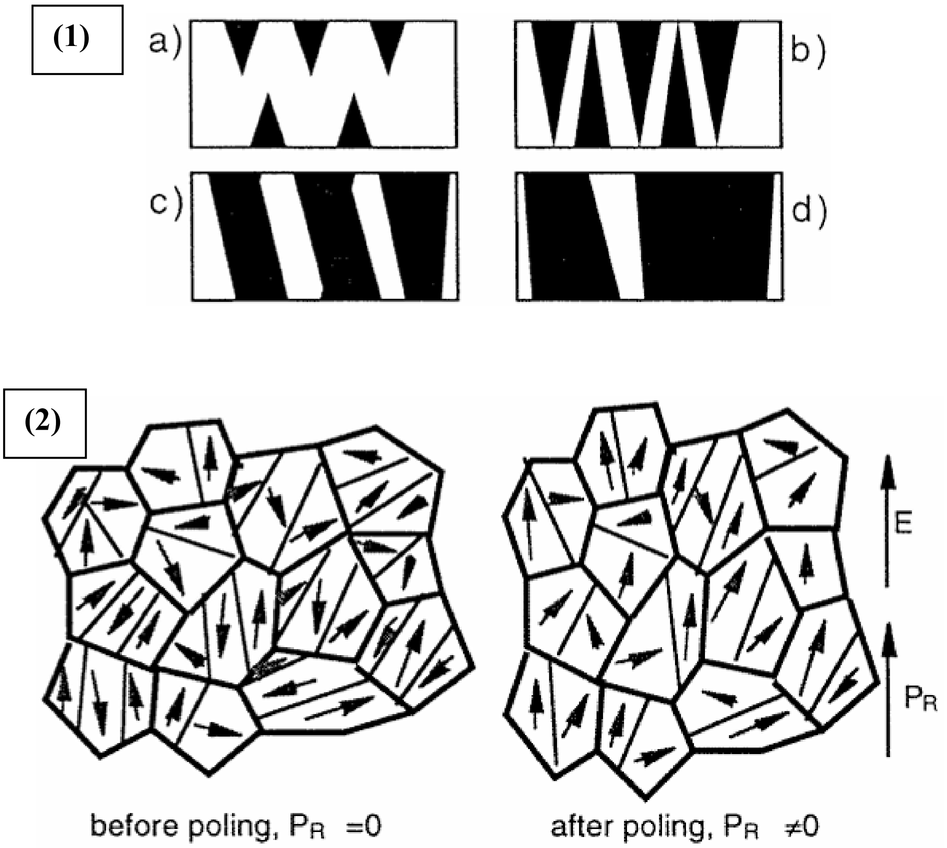


Figure 2.4.3: 1) Probable sequence of polarization switching in ferroelectrics: (a) nucleation of oppositely oriented domains, (b) growth of oppositely oriented domains, (c) sidewise motion of domain walls and (d) coalescence of domains.[46] 2) A polycrystalline ferroelectric with random orientation of grains before and after poling. Many domain walls are present in the poled material, however, the net remanent polarization is nonzero.[27]

2.5 Perovskite Structure and Phase Boundary Engineering

Both KNN and BCHfT are based on the ABO_3 perovskite structure. In this structure, the larger A-site cations occupy the corners of the unit cell, the smaller B-site cation occupies the center of the oxygen octahedron, and oxygen ions form the octahedral framework. In the present KNN–BCHfT system, K^+, Na^+, Ba^{2+}, and Ca^{2+} may be considered A-site-related cations, whereas Nb^{5+}, Ti^{4+}, and Hf^{4+} are B-site-related cations.

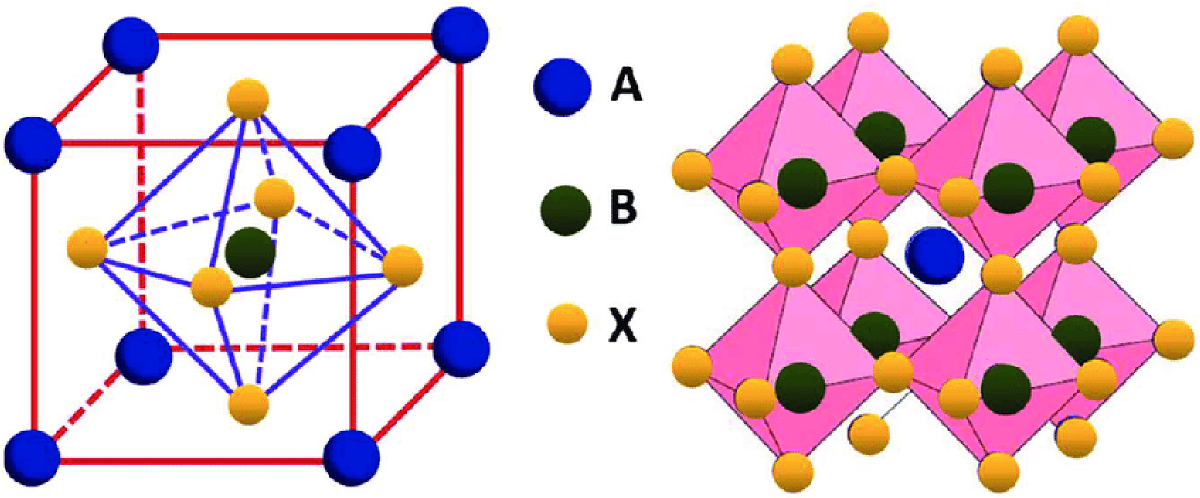


Figure 2.5.1: Perovskite Structure

- **A-site:** big cation (K^+, Na^+, Ba^{2+}, Ca^{2+})
- **B-site:** smaller cation in the octahedral center (Nb^{5+}, Ti^{4+}, Hf^{4+})
- **X-site:** oxygen anions (O^{2-}) — three of them, making the O_3

The stability of the perovskite structure is often discussed using the Goldschmidt tolerance factor:

$$t = \frac{r_A + r_O}{\sqrt{2}(r_B + r_O)} \qquad \textbf{(8)}$$

where r_A, r_B, and r_O are the ionic radii of the A-site cation, B-site cation, and oxygen ion, respectively. When t is close to 1, the structure tends toward a cubic arrangement. When t deviates from 1, lattice distortion occurs, producing tetragonal, orthorhombic, or rhombohedral phases. These distorted phases are essential for ferroelectricity because they create non-centrosymmetric structures and allow spontaneous polarization.[28]

High piezoelectricity in ferroelectric ceramics is strongly related to phase-boundary engineering. In PZT, the morphotropic phase boundary (MPB) is responsible for high piezoelectric performance because different ferroelectric phases coexist, making polarization rotation easier.[29] In KNN-modified piezoceramics, high piezoelectric performance is often associated with polymorphic multistructures rather than a conventional PZT-type MPB; the coexistence of orthorhombic, tetragonal, and/or rhombohedral phases increases the number of possible polarization directions, thereby easing domain rotation and improving piezoelectric response.[30]

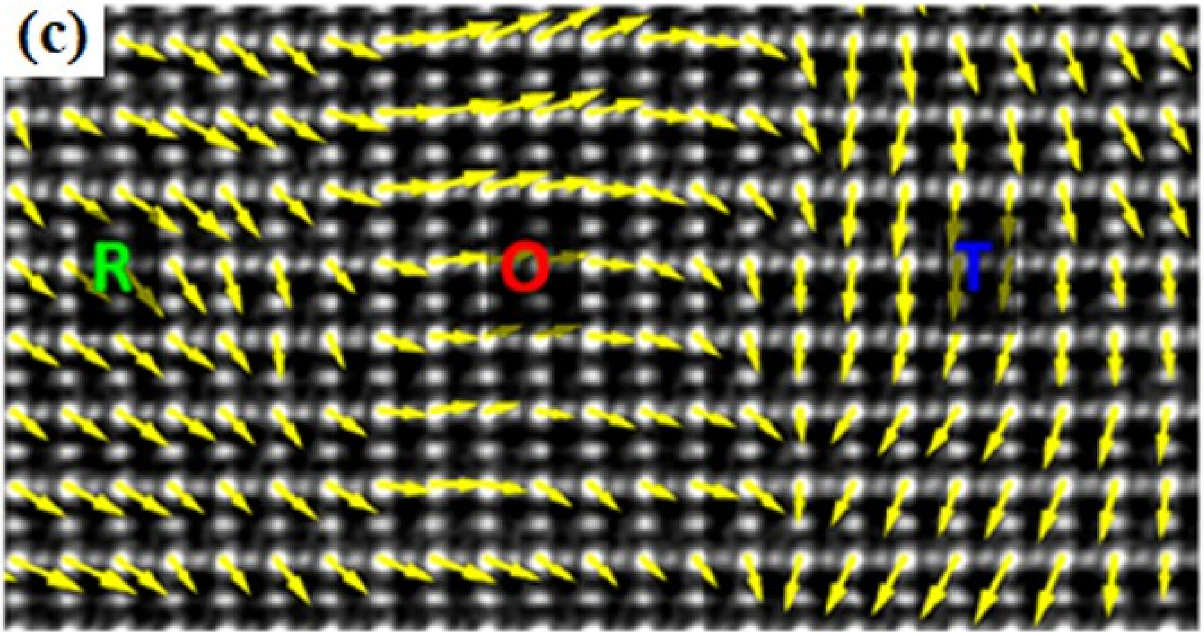


Figure 2.5.2: Atomically resolved contrast-reserved STEM ABFimage along the [110] direction showing polarization rotation from rhombohedral to orthorhombic to tetragonal.[47]

2.6 KNN as a Lead-Free Matrix

$K_{0.5}Na_{0.5}NbO_3$ (KNN) is a lead-free perovskite piezoelectric ceramic formed as a solid solution of ferroelectric $KNbO_3$ and antiferroelectric $NaNbO_3$. It is considered a promising alternative to lead-based piezoceramics because of its high Curie temperature above 400 °C, good ferroelectric properties, and large electromechanical coupling factors.[31] A recent review reports that KNN-based ceramics have an orthorhombic structure at room temperature and phase transition temperatures around $T_{R-O} \approx -110°C$, $T_{O-T} \approx 190°C$, and $T_C \approx 420°C$.[30]

Despite these advantages, pure KNN has several processing difficulties. During high-temperature sintering, potassium and sodium can volatilize, causing deviation from stoichiometry. This can produce A-site vacancies, oxygen vacancies, and secondary phases. These defects can pin domain walls and reduce piezoelectric response. KNN also has a narrow sintering temperature window, so small changes in sintering temperature can cause poor densification, abnormal grain growth, or secondary phase formation.[31]

Therefore, modifying KNN with BCHfT is scientifically reasonable because the modifier may improve densification, reduce harmful microstructural defects, and shift phase transitions toward a more favorable region.

2.7 BCHfT as a $BaTiO_3$-Based Ferroelectric Modifier

$Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$ (BCHT/BCHfT) is a $BaTiO_3$-based lead-free ferroelectric/ piezoceramic composition in which Ca^{2+} substitution at the Ba^{2+} A-site and Hf^{4+} substitution at the Ti^{4+} B-site are used to modify the structural, dielectric, and piezoelectric behavior. Here Hf acts as a B-site dopant, facilitates perovskite-phase formation, produces a denser microstructure with larger grains and lower porosity, sharpens the phase-transition peaks, reduces dielectric loss and coercive field, and increases the piezoelectric d_{33} response of BCHT ceramics.[32]

In BCHT ceramics, Ca^{2+} and Hf^{4+} substitutions play an important role in controlling lattice distortion, phase stability, and electrical properties. Wang et al. reported that the smaller ionic radii of Ca^{2+} and Ti^{4+} compared with Ba^{2+} and Hf^{4+} lead to changes in lattice parameters, unit-cell contraction, and microstrain development. The competition between Ca and Hf contents drives complicated phase evolution and shifts the rhombohedral–orthorhombic, orthorhombic–tetragonal, and Curie transition temperatures. As a result, a ferroelectric orthorhombic–tetragonal phase boundary forms near the optimum composition, giving enhanced dielectric and piezoelectric properties.[33]

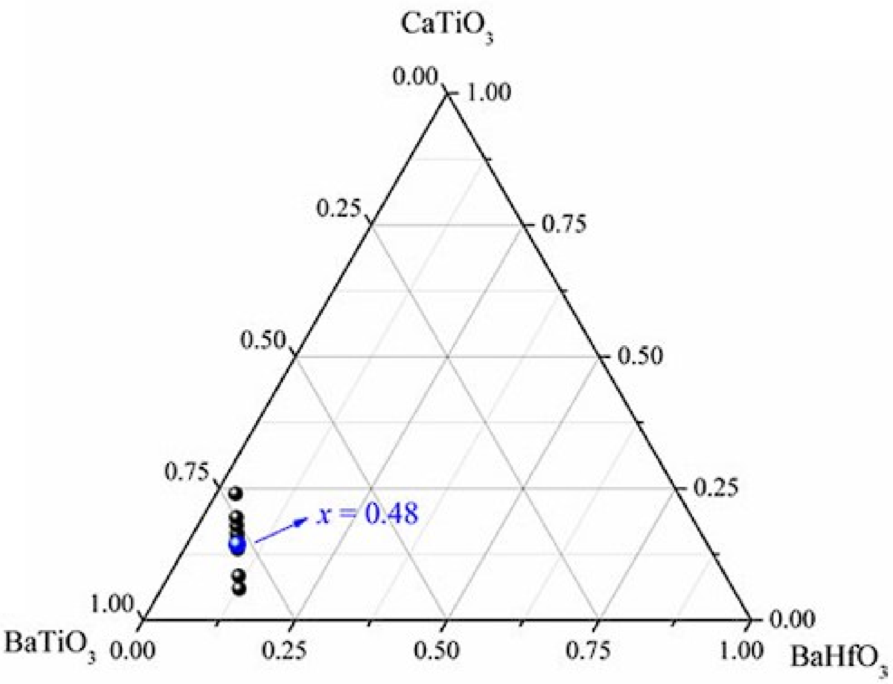


Figure 2.7.1: Schematic representation of the BaTiO3–CaTiO3–BaHfO3 ternary diagram.[33]

Recent work on $Ba_{0.85}Ca_{0.15}Hf_{0.10}Ti_{0.90}O_3$ reported that increasing sintering temperature increased density, grain size, and dielectric constant, and promoted morphotropic behavior involving tetragonal/orthorhombic phase coexistence.[34] This is relevant to the present thesis because the functional properties of BCHfT are not controlled by composition alone; they also depend strongly on sintering temperature, phase constitution, and microstructure. Therefore, when BCHfT is introduced into KNN, its effect should be analyzed not simply as an additive effect, but as a combined structural, microstructural, and dielectric modification effect.

2.8 Rationale for KNN–BCHfT Composite Formation

The main scientific idea of KNN–BCHfT composite formation is to combine the high Curie temperature and lead-free piezoelectric nature of KNN with the phase-tuning and dielectric-modifying ability of BCHfT. Forming a composite between KNN and BCHfT is expected to improve densification, modify microstructure, create favorable phase coexistence, promote domain-wall motion, and enhance stress-to-electric signal conversion.

Modern KNN-based lead-free piezoceramic research is not limited to maximizing the room-temperature piezoelectric coefficient; rather, it aims to achieve a balanced combination of high d_{33}, temperature stability, stable phase-boundary behavior, and controllable polarization response. Additionally atomic-scale local ferroelectric structure design can reduce the local polarization energy barrier and promote polarization rotation in KNN-based ceramics, resulting in a high d_{33} of about 430 pC/N with only ~7% fluctuation from room temperature to 100 °C. After annealing, the temperature-stable range was extended to 150 °C while maintaining d_{33} around 380 pC/N. This indicates that local ferroelectric distortion, phase stability, and polarization-response control are key factors for developing high-performance temperature-stable KNN ceramics.[35]

Similarly, microstructural engineering has become a key approach for improving the temperature stability of KNN-based lead-free piezoceramics, where strategies such as diffuse phase transition with multiphase coexistence, polar nanoregions, domain engineering, multilayer/composition-gradient architectures, atomic-scale local ferroelectric-state design, and defect modulation help stabilize the polymorphic phase transition and suppress temperature-induced structural fluctuations.[36] BCHfT can be understood within this same framework: it may act as a phase-boundary modifier, stress-field modifier, grain-growth controller, and dielectric-stability enhancer.

The most important literature gap is that direct peer-reviewed studies specifically on KNN–BCHfT composites appear much less common than studies on KNN-based systems, BCHT/$BaTiO_3$-based systems, and KNN modified with other perovskite additives. The thesis should be written as a bridging study between KNN phase-boundary engineering and BCHfT/$BaTiO_3$-based microstructure–phase engineering. Therefore, this thesis can contribute by systematically connecting XRD phase formation, SEM microstructure, dielectric behavior, ferroelectric response, and piezoelectric signal conversion in a KNN–BCHfT composite system.

2.9 Defect Chemistry and Alkali Volatilization in KNN-Based Ceramics

Defect chemistry plays a very important role in potassium sodium niobate, $K_{0.5}Na_{0.5}NbO_3$ (KNN), because its structural, microstructural, dielectric, and piezoelectric properties are highly sensitive to compositional stoichiometry. KNN is an ABO_3-type perovskite ceramic in which K^+ and Na^+ occupy the A-site, Nb^{5+} occupies the B-site, and O^{2-} forms the oxygen octahedral framework. Evaporation of Na_2O and K_2O during sintering can prevent the formation of dense, high-performance $Na_{0.5}K_{0.5}NbO_3$ ceramics, while the resulting Na/K loss produces off-stoichiometric vacancy defects. These intrinsic cation and oxygen vacancies influence dielectric relaxation, secondary-phase formation, ion-hopping conduction, domain behavior, dielectric loss, and electromechanical properties.[37]

In KNN ceramics, alkali volatilization mainly produces A-site deficiency. When K or Na is lost from the perovskite lattice during sintering, vacancies are created at the A-site. In Kröger–Vink notation, potassium and sodium vacancies may be represented as V_{K}' and V_{Na}', respectively. Since these vacancies carry an effective negative charge relative to the normal lattice site, charge compensation is required to maintain electrical neutrality. One common compensation mechanism is the formation of oxygen vacancies, $V_{\mathrm{O}}^{\bullet\bullet}$. Therefore, the volatilization of alkali oxides from KNN can be represented schematically as:

$$2\,\mathrm{Na}_{\mathrm{Na}}^{x} + \mathrm{O}_{\mathrm{O}}^{x} \rightarrow \mathrm{Na_2O}(g) + 2V_{\mathrm{Na}}' + V_{\mathrm{O}}^{\bullet\bullet}$$

$$2\,\mathrm{K}_{\mathrm{K}}^{x} + \mathrm{O}_{\mathrm{O}}^{x} \rightarrow \mathrm{K_2O}(g) + 2V_{\mathrm{K}}' + V_{\mathrm{O}}^{\bullet\bullet}$$

These reactions show that the loss of K_2O or Na_2O from the ceramic is not only a simple mass-loss process; it also changes the defect population inside the material.[37] The resulting A-site vacancies and oxygen vacancies may exist as isolated point defects, but they can also associate to form defect complexes such as $(2V_{\mathrm{A}}' - V_{\mathrm{O}}^{\bullet\bullet})^{x}$, where V_A' represents an alkali-site vacancy. Such defect complexes can influence local charge distribution, lattice distortion, grain-boundary behavior, and domain-wall movement. Therefore, alkali volatilization should be considered as a coupled defect-chemistry and microstructure problem rather than only a processing issue.[38]

Oxygen vacancies strongly influence the dielectric and conduction behavior of KNN-based ceramics, especially at elevated temperatures. The high-temperature dielectric relaxation in $0.95K_{0.5}Na_{0.5}NbO_3$–$0.05BaZrO_3$ ceramics is associated with oxygen vacancies, where the short-range hopping of doubly ionized oxygen vacancies contributes to dielectric relaxation, while their long-range motion is responsible for DC electrical conduction. The frequency-dependent dielectric response and impedance behavior therefore indicate that oxygen-vacancy migration and thermally activated charge transport play an important role in the dielectric loss and electrical response of KNN–BZ ceramics.[39]

In the present KNN–BCHfT composite system, defect chemistry becomes even more important because the addition of $Ba_{0.85}Ca_{0.15}Ti_{0.9}Hf_{0.1}O_3$ may influence both the phase structure and the defect equilibrium of KNN. BCHfT contains Ba^{2+} and Ca^{2+} as A-site-related cations and Ti^{4+}

and Hf^{4+} as B-site-related cations. If a small amount of BCHfT dissolves into the KNN lattice, Ba^{2+} or Ca^{2+} may substitute at K^+/Na^+ sites and behave as donor-type substitutions because their valence is higher than that of the alkali cations. Similarly, Ti^{4+} or Hf^{4+} may substitute at Nb^{5+} sites and behave as acceptor-type substitutions because their valence is lower than that of Nb^{5+}. These donor and acceptor type substitutions may partly compensate each other, modify oxygen-vacancy concentration, and change local lattice distortion. However, if BCHfT does not fully dissolve into the KNN lattice, it may remain as a separate phase or form interfaces with the KNN matrix. In that case, the main effect may arise from interfacial polarization, grain-boundary modification, and secondary-phase-related microstructural changes rather than from complete solid-solution defect chemistry.

Such an analysis is very important to understand the dielectric and impedance properties of KNN-BCHfT ceramic materials. A small amount of BCHfT will help increase the polarizability and local distortions of the lattice, improving the ratio of dielectric constant to dielectric loss. Excessive amounts of BCHfT, on the other hand, could facilitate interfacial polarization, defect formation, or heterogeneity in the grain boundary region. This could increase the dielectric loss at low frequencies and cause impedance behaviors that are not ideal. The use of BCHfT cannot only be viewed as an interface modifier; it should also be viewed as a possible defect chemistry modifier.

2.10 Dielectric Properties and Phase Transition Behavior

Dielectric analysis provides important information about phase transitions, defect concentration, dielectric loss, relaxor behavior, and temperature stability. For a normal ferroelectric material above the Curie temperature, dielectric permittivity approximately follows the Curie–Weiss law:

$$\frac{1}{\varepsilon_r} = \frac{T - T_0}{C} \qquad \textbf{(9)}$$

where ε_r is relative permittivity, T is temperature, T_0 is Curie–Weiss temperature, and C is the Curie constant.

For diffuse or relaxor-like ferroelectrics, the modified Curie–Weiss law is often used:

$$\frac{1}{\varepsilon_r} - \frac{1}{\varepsilon_m} = \frac{(T - T_m)^\gamma}{C} \qquad \textbf{(10)}$$

where ε_m is maximum permittivity, T_m is the temperature of maximum permittivity, and γ is the diffuseness parameter. A value of $\gamma = 1$ indicates normal ferroelectric behavior, while $\gamma = 2$ indicates ideal relaxor behavior.[40]

For KNN–BCHfT composites, dielectric measurements as a function of temperature and frequency can reveal whether BCHfT addition broadens the phase transition and improves dielectric stability. A broad dielectric peak is often beneficial for temperature-stable piezoelectric response because it indicates gradual structural change rather than abrupt phase transition. Recent KNN research highlights that limited temperature stability is a major barrier for practical KNN application, and that diffuse phase transition and multiphase coexistence are effective strategies for improving thermal stability.

Dielectric loss, $\tan\delta$, is also important. High dielectric loss may indicate leakage current, oxygen vacancies, poor densification, or conducting grain boundaries.[41] A good KNN–BCHfT composite should ideally show high dielectric permittivity, low dielectric loss, and stable dielectric response over a wide temperature and frequency range.

2.11 Piezoelectric Properties and Stress-to-Electric Signal Conversion

The most direct indicator of stress-to-electric signal conversion is the piezoelectric charge coefficient d_{33}, which can be expressed as:

$$d_{33} = \frac{Q}{F} = \frac{\sigma}{P} \qquad \textbf{(11)}$$

Where σ and P are the corresponding charge density, Q is the generated charge and $F$is the applied mechanical force. A higher d_{33}means that the material can generate more electrical charge under mechanical stress.[42]

Compared with pure KNN ceramics, KNN-modified lead-free piezoceramics can exhibit significantly enhanced piezoelectric properties when polymorphic phase coexistence and microstructural texturing are properly engineered; in particular, O–T multistructure ceramics have been reported to show d_{33} values of 220–502 pC/N, while R–T multistructure ceramics can reach approximately 430–550 pC/N because the coexistence of multiple ferroelectric structures facilitates easier polarization rotation and domain switching.[43]

For the present KNN–BCHfT thesis, the expected improvement in d_{33}can be explained by four linked mechanisms:

1. First, BCHfT may shift KNN phase boundaries toward room temperature, increasing polarization rotation.
2. Second, BCHfT may improve densification and reduce pores, allowing better poling.
3. Third, Hf-related lattice distortion may create local structural heterogeneity and broaden phase transitions.
4. Fourth, improved microstructure may increase reversible domain-wall motion and reduce defect pinning.

These mechanisms agree with the uploaded thesis goal of forming a stable lead-free composite, optimizing phase boundaries, improving densification, increasing domain-wall mobility, and enhancing dielectric/polarization stability.

2.12 Literature Gap and Research Significance

The literature clearly shows that KNN-based ceramics are among the most promising lead-free piezoelectric systems. However, pure KNN suffers from processing difficulty, secondary phase formation, poor densification, and temperature-sensitive piezoelectric response. The literature also shows that $BaTiO_3$-based systems such as BCHT/BCHfT can be useful for phase tuning, dielectric modification, and microstructural control. However, direct studies on KNN-BCHfT composites are still much less developed than studies on KNN-only systems, BCHT/BCZT systems, and other KNN-based modified ceramics.

Therefore, the present thesis is significant because it attempts to establish a structure–property relationship in a lead-free KNN-BCHfT composite system. The expected contribution is not only to show whether d_{33} increases, but also to explain why it increases or decreases by connecting XRD phase formation, SEM/EDS microstructure, dielectric behavior, and piezoelectric response. This approach is consistent with recent KNN research trends, where the strongest studies focus on the interaction between phase structure, local distortion, microstructure, defects, domain-wall motion, and temperature stability rather than on composition alone.

Chapter:3 - Methodology

3.1 Experimental Overview

In this work, lead-free piezoelectric ceramic powder of potassium sodium niobate, $K_{0.5}Na_{0.5}NbO_3$, abbreviated as KNN, was synthesized by the conventional solid-state reaction method. To improve the piezoelectric response, densification, grain morphology, and electromechanical conversion behavior, KNN was further combined with $Ba_{0.85}Ca_{0.15}Ti_{0.9}Hf_{0.1}O_3$, abbreviated as BCHfT. Pure KNN and pure BCHfT powders were first synthesized separately. Then, the calcined powders were mixed in selected weight percentages to prepare KNN–BCHfT composite ceramics.

The prepared compositions were:

KNN–1 wt% BCHfT
KNN–1.5 wt% BCHfT

3.2 Raw Materials

The raw materials used for the synthesis of KNN are listed in Table 3.1.

Chemical name	Chemical formula	Purity	Molecular mass, g/mol	Manufacturer	Purpose
Niobium pentoxide	Nb_2O_5	99%	265.81	Research Lab India	Nb source
Sodium carbonate	Na_2CO_3	99%	105.95	Merck Specialties Pvt. Ltd., India	Na source
Potassium carbonate	K_2CO_3	99%	138.21	Merck Specialties Pvt. Ltd., India	K source

Table 3.2.1: Raw Materials for KNN

For BCHfT synthesis, the following raw materials should be listed using the exact bottle information from your lab:

Chemical name	Chemical formula	Purity	Molecular mass, g/mol	Manufacturer	Purpose
Barium carbonate	$BaCO_3$	99%	197.34	Merck Specialties Pvt. Ltd., India	Ba source

Calcium carbonate	$CaCO_3$	99%	100.0869	Merck Specialties Pvt. Ltd., India	Ca source
Titanium dioxide	TiO_2	99%	79.866	Merck Specialties Pvt. Ltd., India	Ti source
Hafnium dioxide	HfO_2	98%	210.49	Sigma-Aldrich	Hf source

Table 3.2. 2: Raw Materials for BCHfT

3.3 Stoichiometric Calculation for KNN

The target composition of KNN was:

$K_{0.5}Na_{0.5}NbO_3$

The molecular weight of KNN was calculated as:

$$M_{KNN} = (39 \times 0.5) + (23 \times 0.5) + 92.90 + (16 \times 3)$$
$$M_{KNN} = 171.9 \text{ g/mol}$$

For the synthesis of a 5 g batch of KNN, the required amount of each precursor was calculated by considering the stoichiometric coefficient of each element, molecular mass of the precursor, and purity of the chemical.

The balanced precursor relation may be expressed as:

$$0.25K_2CO_3 + 0.25Na_2CO_3 + 0.5Nb_2O_5 \rightarrow K_{0.5}Na_{0.5}NbO_3 + CO_2$$

The general equation used for precursor calculation was:

$$m_i = \frac{m_{target}}{M_{KNN}} \times \frac{\nu_i M_i}{P_i} \qquad \textbf{(12)}$$

where,
m_i= required mass of precursor,
m_{target}= target mass of KNN powder,
M_{KNN}= molecular weight of KNN,
ν_i= stoichiometric coefficient of precursor,
M_i= molecular weight of precursor,
P_i= purity fraction of precursor.

Calculation for Nb_2O_5

$$Nb_2O_5 = \frac{92.90 \times 5}{171.9 \times 0.99} \times \frac{265.81}{92.90 \times 2}$$
$$Nb_2O_5 = 3.9052 \text{ g}$$

Calculation for K_2CO_3

$$K_2CO_3 = \frac{39 \times 0.5 \times 5}{171.9 \times 0.99} \times \frac{138.21}{39 \times 2}$$
$$K_2CO_3 = 1.0152 \text{ g}$$

Calculation for Na_2CO_3

$$Na_2CO_3 = \frac{23 \times 0.5 \times 5}{171.9 \times 0.99} \times \frac{105.95}{23 \times 2}$$
$$Na_2CO_3 = 0.7782 \text{ g}$$

For a 5 g KNN batch with 99% purity raw materials, the calculated masses were:

Precursor	Calculation basis	Required mass
Nb_2O_5	Nb source	3.9052 g
K_2CO_3	K source	1.0152 g
Na_2CO_3	Na source	0.7782 g

Table 3.3.1: Precursor needed for KNN preparation

Although the final target ceramic mass was 5 g, the total weighed precursor mass was higher because carbonate precursors release CO_2 during calcination.

3.4 Stoichiometric Calculation for BCHfT

The target BCHfT composition was:

$Ba_{0.85}Ca_{0.15}Ti_{0.9}Hf_{0.1}O_3$

The balanced precursor relation may be expressed as:

$0.85BaCO_3+0.15CaCO_3+0.90TiO_2+0.10HfO_2 \rightarrow Ba_{0.85}Ca_{0.15}Ti_{0.9}Hf_{0.1}O_3+CO_2$

The molecular weight of BCHfT was calculated as:

M(BCHfT) = (137.327 × 0.85) + (40.078 × 0.15) + (47.867 × 0.9) + (178.49 × 0.1) + (16 × 3)
M(BCHfT) = 231.66895 g/mol

Calculation for $BaCO_3$

$$BaCO_3 = \frac{197.34 \times 5 \times 0.85}{231.66895 \times 0.99}$$
$$BaCO_3 = 3.6568\ \mathrm{g}$$

Calculation for $CaCO_3$

$$CaCO_3 = \frac{100.0869 \times 0.15 \times 5}{231.66895 \times 0.99}$$
$$CaCO_3 = 0.3273\ \mathrm{g}$$

Calculation for TiO_2

$$TiO_2 = \frac{79.866 \times 0.9 \times 5}{231.66895 \times 0.99}$$
$$TiO_2 = 1.5670\ \mathrm{g}$$

Calculation for HfO_2

$$HfO_2 = \frac{210.49 \times 0.1 \times 5}{231.66895 \times 0.98}$$
$$HfO_2 = 0.4636\ \mathrm{g}$$

For preparing a 5 g batch of BCHfT powder, the required precursor amounts were:

Raw material	Required amount
$BaCO_3$	3.6568 g
$CaCO_3$	0.3273 g
TiO_2	1.5670 g
HfO_2	0.4636 g

Table 3.4.1: Precursor needed for BCHfT preparation

The total weighed precursor mass was higher than 5 g because $BaCO_3$ and $CaCO_3$ release CO_2 during calcination.

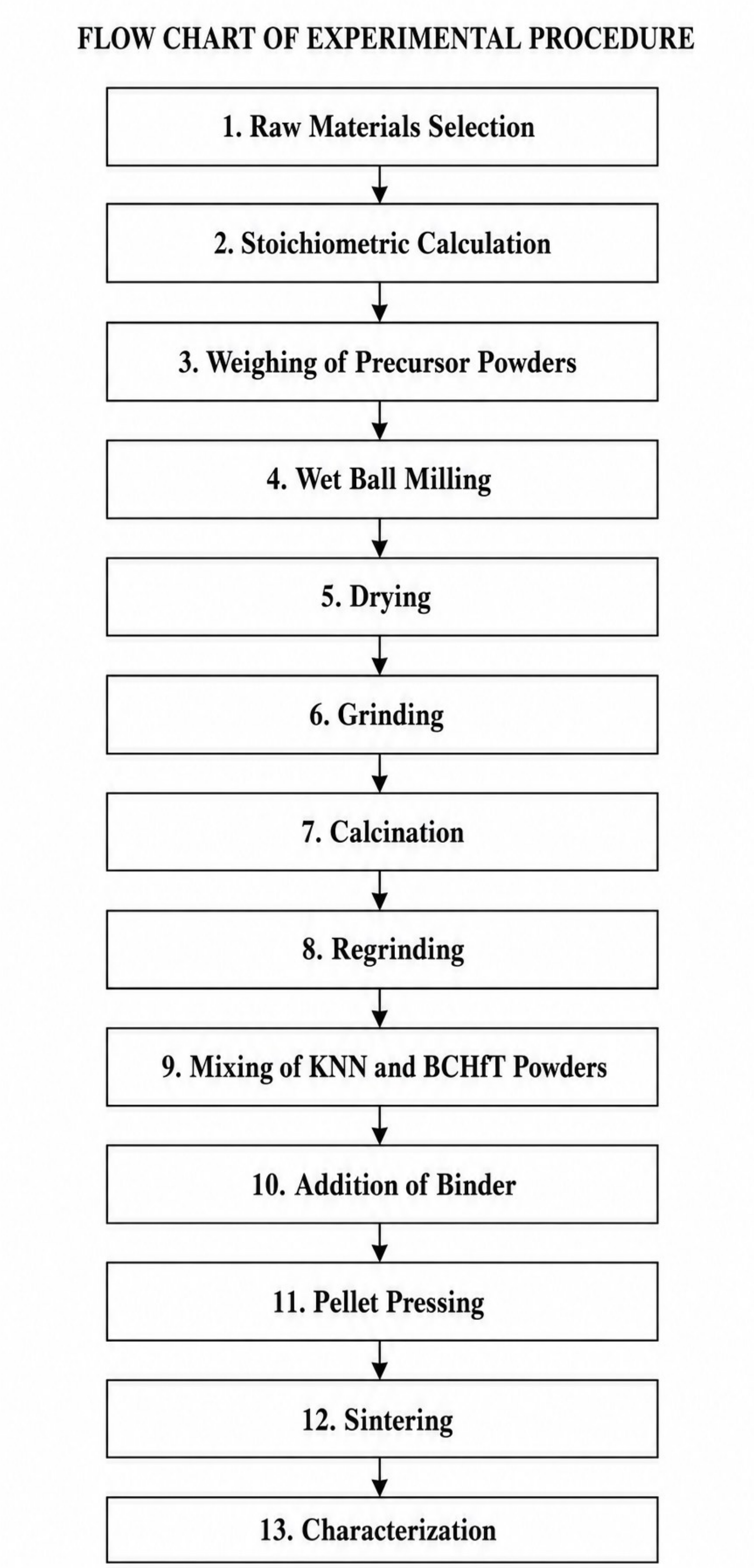


Figure 3.4.1: Synthesis Route of KNN and BCHfT

3.5 Synthesis of KNN Powder

The calculated amounts of Nb_2O_5, K_2CO_3, and Na_2CO_3 were carefully weighed using an analytical balance. The powders were then transferred into a zirconia milling jar for wet ball milling. Zirconia balls were used as the milling media, and the ball-to-powder weight ratio was maintained at approximately 5:1. Ethanol was added as the milling medium until about two-thirds of the jar volume was filled.

The powder mixture was ball milled for 24 hours to ensure proper mixing and compositional homogeneity. After milling, the slurry was collected and dried in an oven at approximately 70°C for 24 hours to remove ethanol completely.

The dried powder was then crushed and ground using an agate mortar and pestle to break soft agglomerates. The ground powder was placed in an alumina crucible and calcined at 950°C for 6 hours. A heating rate of about 5°C/min was used, followed by furnace cooling to room temperature. The calcination step was performed to promote solid-state reaction among the precursor oxides/carbonates and to form the KNN perovskite phase.

After calcination, the powder was again ground thoroughly to obtain fine and uniform KNN powder.

3.6 Synthesis of BCHfT Powder

The calculated amounts of $BaCO_3$, $CaCO_3$, TiO_2, and HfO_2 were weighed according to the stoichiometric ratio of $Ba_{0.85}Ca_{0.15}Ti_{0.9}Hf_{0.1}O_3$. The weighed powders were transferred into a zirconia milling jar. Zirconia balls were used as the milling media with a ball-to-powder weight ratio of approximately 5:1. Ethanol was used as the milling medium and filled about two-thirds of the jar.

The powder mixture was wet ball milled for 24 hours to ensure proper homogeneity. After milling, the slurry was dried in an oven at 80°C for 24 hours. The dried powder was ground using a mortar and pestle.

The ground BCHfT precursor powder was placed in an alumina crucible and calcined at 950°C for 6 hours with a heating rate of 5°C/min. After calcination, furnace cooling was carried out to room temperature. The calcined BCHfT powder was then reground to obtain fine and homogeneous powder.

3.7 Preparation of KNN–BCHfT Composite Powder

The calcined KNN and BCHfT powders were mixed according to the selected compositions. For KNN–1 wt% BCHfT, 4.95 g of calcined KNN powder was mixed with 0.05 g of calcined BCHfT powder. For KNN–1.5 wt% BCHfT, 4.925 g of calcined KNN powder was mixed with 0.075 g of calcined BCHfT powder.

The powder mixture was mixed thoroughly to ensure uniform distribution of BCHfT within the KNN matrix. Homogeneous mixing is important because the distribution of BCHfT can influence densification, grain growth, dielectric behavior, and piezoelectric performance of the final ceramic composite.

For a 5 g batch:

KNN–1 wt% BCHfT

Component	Required amount
KNN powder	4.95 g
BCHfT powder	0.05 g
Total	5.00 g

Table 3.7.1: KNN–1 wt% BCHfT

KNN–1.5 wt% BCHfT

Component	Required amount
KNN powder	4.925 g
BCHfT powder	0.075 g
Total	5.00 g

Table 3.7.2: KNN–1.5 wt% BCHfT

3.8 Binder Addition and Pellet Preparation

For pellet preparation, 5 wt% polyvinyl alcohol (5gm PVA in 100mL Deionized water), PVA, binder solution was added to the composite powder. The powder and binder were mixed and ground using a mortar and pestle for approximately 30–40 minutes to ensure uniform binder distribution.

For each pellet, approximately 1.2 g of powder was taken and placed into a circular die. The powder was pressed into pellet form using a hydraulic press with 5kN force. The prepared green pellets were dried before sintering.

3.9 Sintering of Pellets

The prepared pellets were sintered at 1090°C for 10 hours. A heating rate of 5°C/min was used, followed by furnace cooling to room temperature. The sintering process was carried out to improve densification, reduce porosity, and develop ceramic grains with better mechanical and electrical properties.

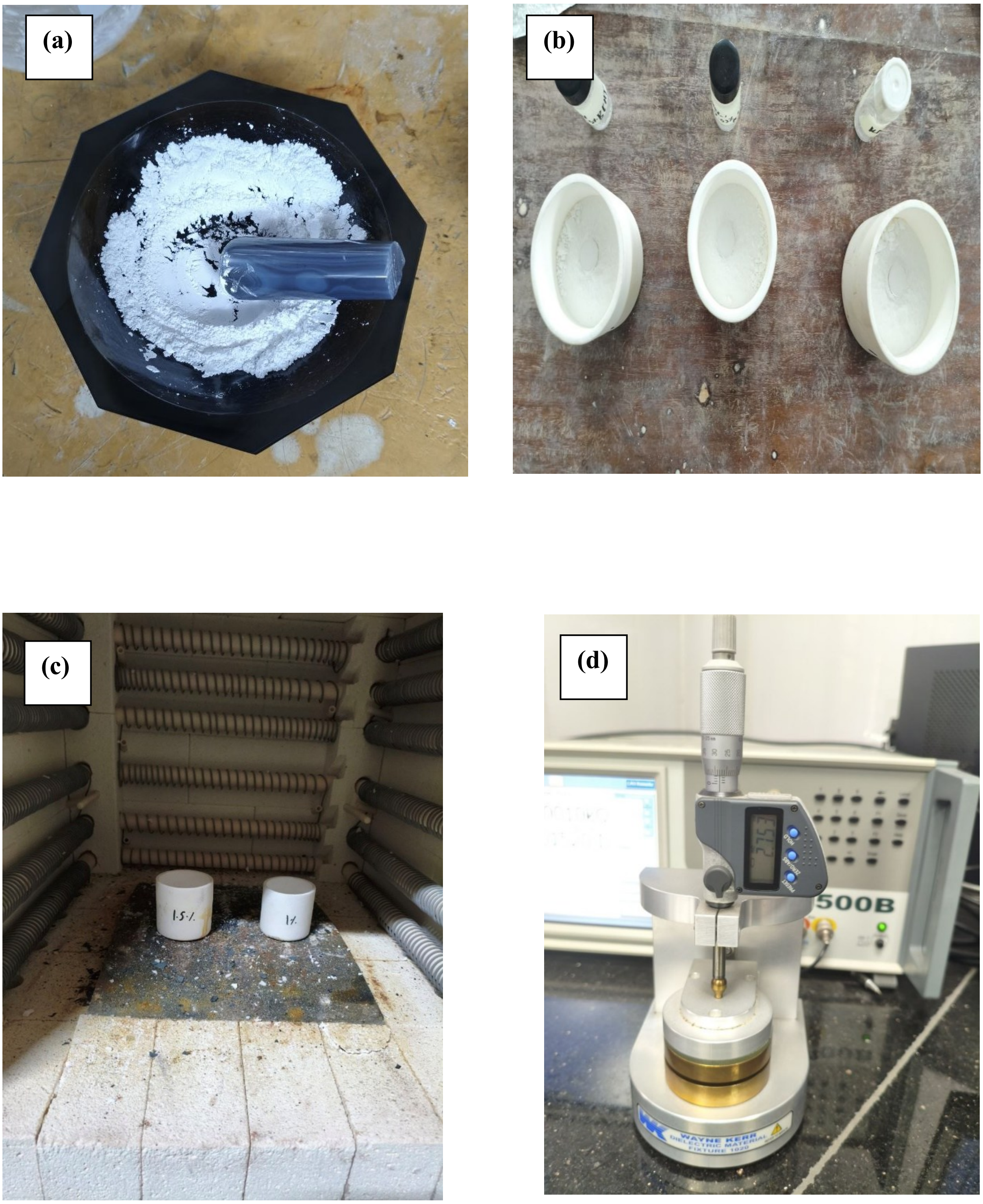


Figure 3.9.1: Experimental processing and characterization steps: (a) grinding of powder samples using mortar and pestle, (b) compacted green pellets before sintering, (c) pellets loaded into the furnace for sintering, and (d) frequency-dependent dielectric measurement setup.

Chapter:4 - Characterization

4.1 Introduction

Characterization is an essential part of ceramic materials characterization research, which helps identify the characteristics of their structure, microstructure, composition, dielectric, and piezoelectric properties. In the current thesis, characterization of synthesized KNN-BCHfT ceramic material samples was performed through various experimental methods. XRD analysis was used to establish the crystalline structure of the synthesized samples, while SEM analysis was employed to analyze the morphology and microstructure of the samples. Energy Dispersive X-ray Spectroscopy was used to confirm the presence and distribution of constituent elements in the prepared ceramics.

In addition, dielectric measurements were carried out as a function of frequency and temperature to study the polarization behavior, dielectric constant, dielectric loss, relaxation behavior, and phase transition characteristics of the samples. Finally, piezoelectric testing was performed to evaluate the electromechanical response of the ceramics after poling.

4.2 X Ray Diffraction (XRD)

A popular method in many engineering applications for determining a material's crystalline structure is X-ray diffraction (XRD). The prepared pristine powders and piezoelectric ceramics were characterized via an EMPYREAN diffractor system to find out their crystalline structure and understand their properties, including crystallinity and grain size. These properties significantly affect the characteristics of materials.

The configuration and parameters of the used EMPYREAN diffractor system is given in the Table:

Parameter	**Value**
X-ray generator	40 mA, 45 kV
Scan type	Continuous
PSD Mode	Scanning
PSD Length (°2θ)	3.35
Divergence Slit Type	Fixed
Divergence Slit Size (°)	0.7197
Anode Material	Cu
$K_{\alpha 1}$ wavelength (Å)	1.54060

$K_{\alpha 2}$ wavelength (Å)	1.54443
$K_{\beta 1}$ wavelength (Å)	1.39225
$K_{\beta 2}$ wavelength (Å)	1.38113
K Absorption Edge	1.37868
K_{β} filter material	Ni
K_{β} filter thickness	0.02 mm
Scan axis	Gonio
Scan range	5 ~ 80°

Table 4.2.1: EMPYREAN XRD System Parameter

4.2.1 Diffraction Parameter

To calculate crystallite size and micro-strain following formulas were used:

Crystallite Size:

$$L = \frac{K\lambda}{(\beta \cos \theta)} \text{ .... } \mathbf{(13)}$$

Micro-strain:

$$\varepsilon = \frac{\beta}{(4 \tan \theta)} \text{ .... } \mathbf{(14)}$$

Here, λ = X-ray wavelength (0.15406 nm), θ = diffraction angle (radian), β = FWHM at diffraction peak (full-width half maxima) (radian).

4.3 Scaning Electron Microscopy (SEM)

Using FESEM (JEOL JSM-7600F), an applied voltage of 5 kV and a working distance of 7.4 mm, the morphological investigation of as-synthesized KNN-B VAFO was performed. Images were captured under 10000x, 20000x, and 30000x among other magnitudes.

4.4 Dielectric Characterization

Dielectric characterization was carried out to investigate the electrical response, polarization behavior, dielectric loss, AC conductivity, and impedance behavior of the prepared ceramic samples. In the present work, both frequency-dependent and temperature-dependent dielectric measurements were performed. The measurements were conducted using a Wayne Kerr 6500B

Series Precision Impedance Analyzer connected with a Wayne Kerr 1020 Material Test Fixture. For temperature-dependent measurement, the same impedance measurement system was coupled with a controlled heating arrangement so that the dielectric response of the ceramic pellets could be recorded at different temperatures.

Before measurement, both surfaces of the sintered ceramic pellets were polished properly and coated with platinum electrodes to ensure good electrical contact. The sample thickness and diameter were measured before dielectric testing. The sample thickness and diameter was:

Specimen	Thickness (mm)	Diameter (mm)
KNN	2.78	12.1
BCHfT	3.1	11.7
KNN-1%BCHfT	1.8	11.97
KNN-1.5%BCHfT	2.8	11.8

Table 4.4.1: Specimen thickness and diameter.

The corresponding electrode area was calculated from the circular pellet geometry.

The dielectric measurement system recorded the capacitance, dissipation factor, impedance, resistance, and reactance of the samples. These measured quantities were then used to calculate the real part of dielectric permittivity, imaginary part of dielectric permittivity, AC conductivity, electric modulus, and impedance parameters.

4.4.1 Measurement Setup and Parameters

The dielectric measurements were performed using a parallel equivalent circuit model. The applied AC voltage was 1 V and the DC bias was kept off during measurement. The frequency-dependent dielectric data were collected by sweeping the frequency from 20 Hz to 1 MHz. The temperature-dependent dielectric data were collected over the temperature range of 30°C to 540°C at selected frequencies of 100 Hz, 1 kHz, 10 kHz, and 1 MHz.

The main measurement parameters used in this study are listed in Table below:

Parameter	**Value**
Instrument	Wayne Kerr 6500B Series Precision Impedance Analyzer
Fixture	Wayne Kerr 1020 Material Test Fixture
Equivalent circuit	Parallel
AC voltage	1 V
DC bias	Off

Frequency range for frequency-dependent test	20 Hz – 1 MHz
Frequencies for temperature-dependent test	100 Hz, 1 kHz, 10 kHz, 1 MHz
Temperature range	30°C – 540°C
Sample thickness	1.83 mm
Sample diameter	11.83 mm
Electrode material	Platinum
Measured parameters	C, D/tanδ, Z, R, X, phase angle
Calculated parameters	ε', ε'', σac, M', M'', Z', and $-Z''$

Table 4.4.2: Dielectric measurement system and experimental parameters

4.4.2 Frequency-Dependent Dielectric Measurement

Frequency-dependent dielectric measurements were performed to study the variation of dielectric properties with applied frequency. The measurements were carried out for KNN, BCHfT, KNN–1% BCHfT, and KNN–1.5% BCHfT ceramic samples. During the measurement, the frequency was varied from 20 Hz to 1 MHz while the AC voltage was kept constant at 1 V.

The impedance analyzer recorded the capacitance, dissipation factor, impedance, phase angle, and reactance of the samples. From these measured values, the real dielectric constant, imaginary dielectric constant, AC conductivity, and Nyquist plots were obtained.

4.4.3 Temperature-Dependent Dielectric Measurement

Temperature-dependent dielectric measurements were performed to observe the thermal stability and phase transition behavior of the prepared ceramic samples. The dielectric properties were measured from 30°C to 540°C at selected frequencies of 100 Hz, 1 kHz, 10 kHz, and 1 MHz.

During temperature-dependent measurement, the sample was placed between the electrodes of the material test fixture and connected to the impedance analyzer. The temperature was gradually increased using the heating arrangement, and the corresponding dielectric data were recorded at each selected temperature.

4.4.4 Equations Used for Dielectric Calculation

The electrode area of the circular pellet was calculated using the following equation:

$A = \frac{\pi d^2}{4}$ **(15)**

where A is the electrode area and d is the diameter of the pellet.

The real part of the relative dielectric permittivity was calculated from the measured capacitance using the relation:

$\varepsilon' = \frac{Ct}{\varepsilon 0 A}$ **(16)**

where ε' is the real dielectric permittivity, C is the measured capacitance, t is the sample thickness, ε_0 is the permittivity of free space, and A is the electrode area.

The imaginary part of the dielectric permittivity was calculated using:

$\varepsilon'' = \varepsilon' \tan\delta$ **(17)**

where ε'' is the imaginary dielectric permittivity and $\tan\delta$ is the dielectric loss tangent or dissipation factor.

The angular frequency was calculated as:

$\omega = 2\pi f$ **(18)**

where ω is the angular frequency and f is the applied frequency.

The AC conductivity was calculated using the equation:

$\sigma_{ac} = \omega\varepsilon_0\varepsilon'\tan\delta$ **(19)**

or,

$\sigma_{ac} = 2\pi f\varepsilon_0\varepsilon'\tan\delta$ **(20)**

where σac is the AC conductivity, f is the frequency, ε_0 is the permittivity of free space, ε' is the real dielectric permittivity, and $\tan\delta$ is the dielectric loss tangent.

For impedance analysis, the complex impedance can be expressed as:

$Z^* = Z' - jZ''$ **(21)**

where Z' is the real part of impedance and Z'' is the imaginary part of impedance. The Nyquist plot was obtained by plotting Z' along the x-axis and $-Z''$ along the y-axis.

For the parallel equivalent circuit, the susceptance and conductance were calculated as:

$B = \omega C$ **(22)**

$G = D\omega C$ **(23)**

where B is susceptance, G is conductance, C is capacitance, and D is the dissipation factor.

The real and imaginary components of impedance were calculated using:

$Z' = \frac{G}{(G^2 + B^2)}$ **(24)**

$$-Z'' = \frac{B}{(G^2 + B^2)} \qquad \textbf{(25)}$$

The electric modulus formalism was also used to study the relaxation behavior of the samples. The complex electric modulus is expressed as:

$$M^* = \frac{1}{\varepsilon^*} \qquad \textbf{(26)}$$

The real and imaginary parts of the electric modulus were calculated as:

$$M' = \frac{\varepsilon'}{(\varepsilon'^2 + \varepsilon''^2)} \qquad \textbf{(27)}$$

$$M'' = \frac{\varepsilon''}{(\varepsilon'^2 + \varepsilon''^2)} \qquad \textbf{(28)}$$

where M′ is the real electric modulus and M″ is the imaginary electric modulus.

The activation energy for conduction can be estimated from the Arrhenius relation:

$$\sigma_{ac} = \sigma_0 \exp\left(\frac{-Ea}{kBT}\right) \qquad \textbf{(29)}$$

Taking natural logarithm,

$$\ln\sigma_{ac} = \ln\sigma_0 - \frac{Ea}{kBT} \qquad \textbf{(30)}$$

where σ_0 is the pre-exponential factor, Ea is the activation energy, k_B is the Boltzmann constant, and T is the absolute temperature in Kelvin. When $\ln\sigma_{ac}$ is plotted against 1000/T, the activation energy can be calculated from the slope using:

$$E_a = -\text{slope} \times k_B \times 1000 \qquad \textbf{(31)}$$

4.5 Piezoelectric Characterization

Piezoelectric characterization was carried out to evaluate the electromechanical performance of the prepared ceramic samples. Before piezoelectric measurement, the sintered and electroded pellets were poled under a DC electric field of 5kV/mm at 120^0C for 20mins. After poling, the samples were aged for 24 hours before measurement.

The piezoelectric coefficient d33 was measured using a SINOCERA d33 meter. The d33 value represents the electric charge generated per unit mechanical force applied along the poling direction. A higher d33 value indicates better piezoelectric response and stronger electromechanical coupling.

Chapter: 5 – Result & Discussion

5.1 XRD Analysis:

5.1.1 Pure KNN ($K_{0.5}Na_{0.5}NbO_3$)

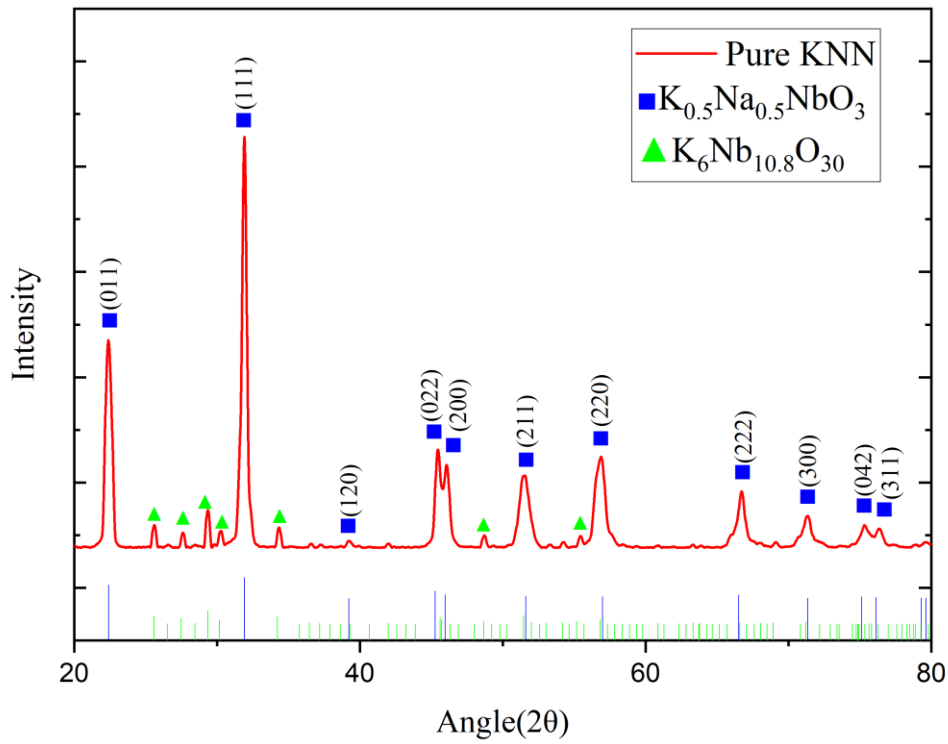


Figure 5.1.1: XRD peaks of pure KNN.

The diffraction pattern of pure KNN shows several sharp and well-defined peaks, indicating good crystallinity of the prepared ceramic. The major diffraction peaks can be assigned mainly to the potassium sodium niobate phase, $K_{0.5}Na_{0.5}NbO_3$, which matches well with the standard JCPDS/PDF card no. 01-085-7128. This reference card identifies KNN as an orthorhombic phase with space group Amm2. Therefore, the presence of the KNN perovskite phase is confirmed in the pure KNN sample.

In addition to the main KNN phase, some extra diffraction peaks are also detected in the pure KNN pattern. These secondary reflections can be indexed to a potassium niobium oxide phase, corresponding to JCPDS/PDF card no. 01-070-5051. This card lists the phase as potassium niobium oxide, $K_6Nb_{10.8}O_{30}$, with a tetragonal crystal structure and space group P4/mbm.

5.1.2 Pure BCHfT ($Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$)

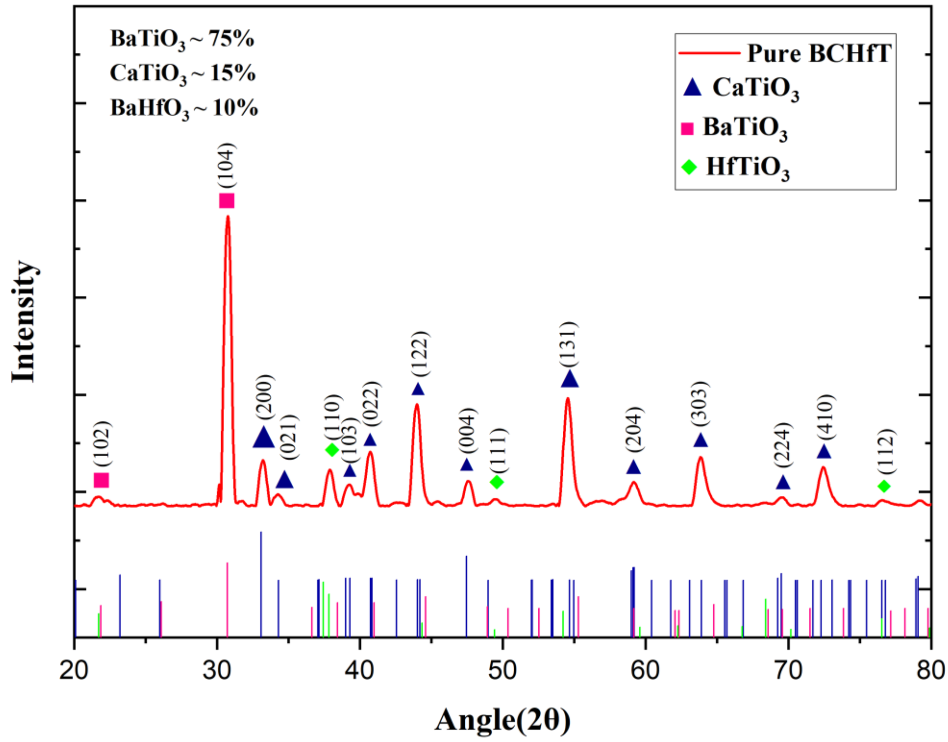


Figure 5.1.2: XRD peaks of pure BCHfT.

For the pure BCHfT ceramic, the diffraction peaks can be indexed with barium titanium oxide, calcium titanium oxide, and hafnium titanium oxide phases. The $BaTiO_3$-related phase is matched with JCPDS/PDF card no. 00-013-0379, which is listed as barium titanium oxide with a hexagonal crystal structure. The $CaTiO_3$ phase is matched with JCPDS/PDF card no. 01-070-8503, corresponding to calcium titanium oxide with an orthorhombic perovskite structure and space group Pbnm. The Hf–Ti–O phase is matched with JCPDS/PDF card no. 00-034-0884, assigned to hafnium titanium oxide, $HfTiO_3$, with a hexagonal structure and space group P6/mmm.

Sample	Identified phase	Chemical formula	JCPDS/PDF card no.	Crystal structure / space group
Pure KNN	Potassium sodium niobate / KNN	$(K_{0.5}Na_{0.5})NbO_3$	01-085-7128	Orthorhombic, Amm2
Pure KNN	Potassium niobium oxide / KNbO-related secondary phase	$K_6Nb_{10.8}O_{30}$	01-070-5051	Tetragonal, P4/mbm
Pure BCHfT	Barium titanium oxide	$Ba(Ti_{0.48}Ti_{0.52})O_{2.76}$	00-013-0379	Hexagonal
Pure BCHfT	Calcium titanium oxide / calcium titanate	$CaTiO_3$	01-070-8503	Orthorhombic, Pbnm
Pure BCHfT	Hafnium titanium oxide	$HfTiO_3$	00-034-0884	Hexagonal, P6/mmm

Table 5.1.1: Phase identification and key XRD features of pure KNN and pure BCHfT ceramics.

5.1.3 Composite KNN-BCHfT

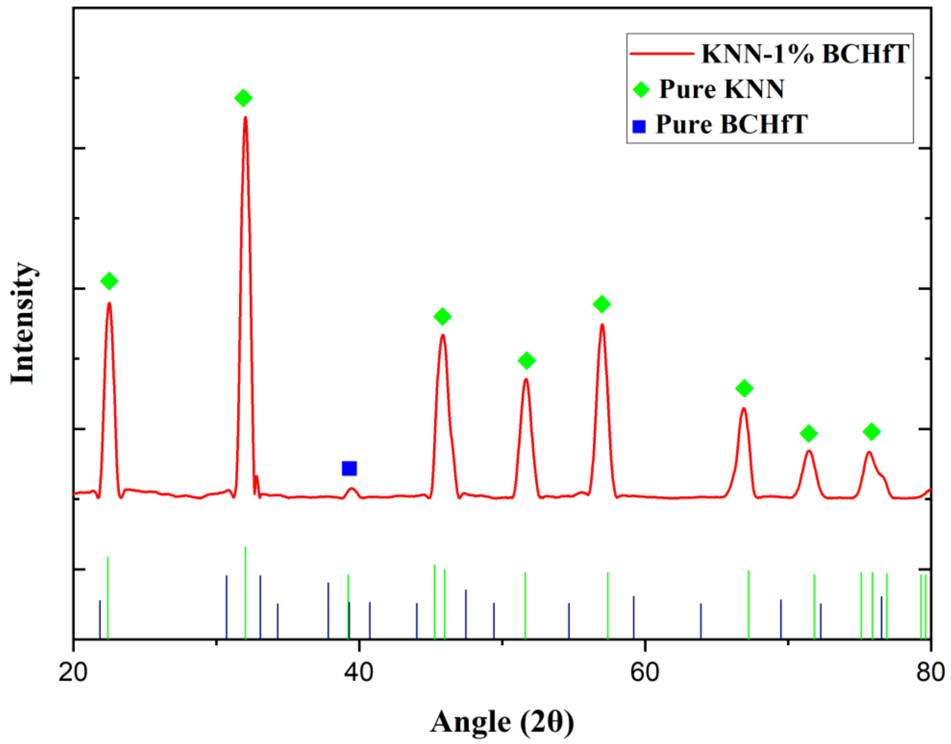


Figure 5.1.3: XRD Analysis of KNN-1% BCHfT

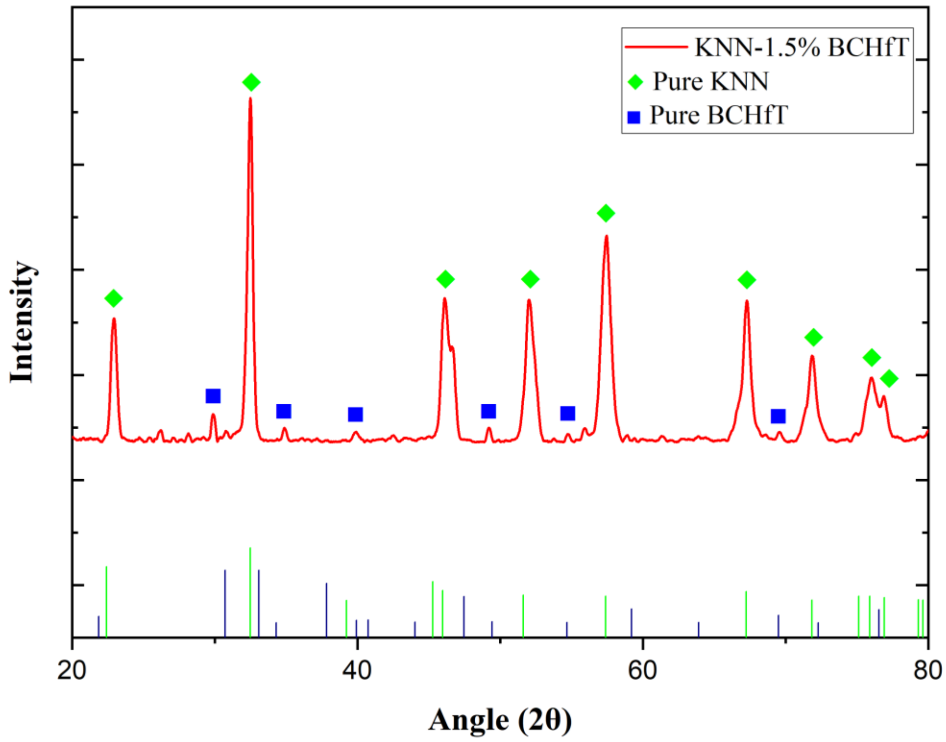


Figure 5.1.4: XRD Analysis of KNN-1.5% BCHfT

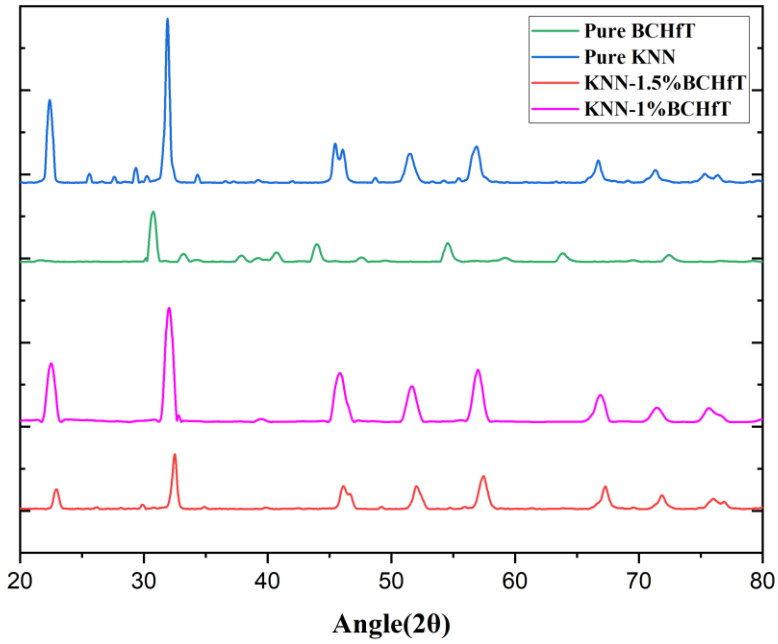


Figure 5.1.5: Stacking all XRD curves together.

5.2 Dielectric test

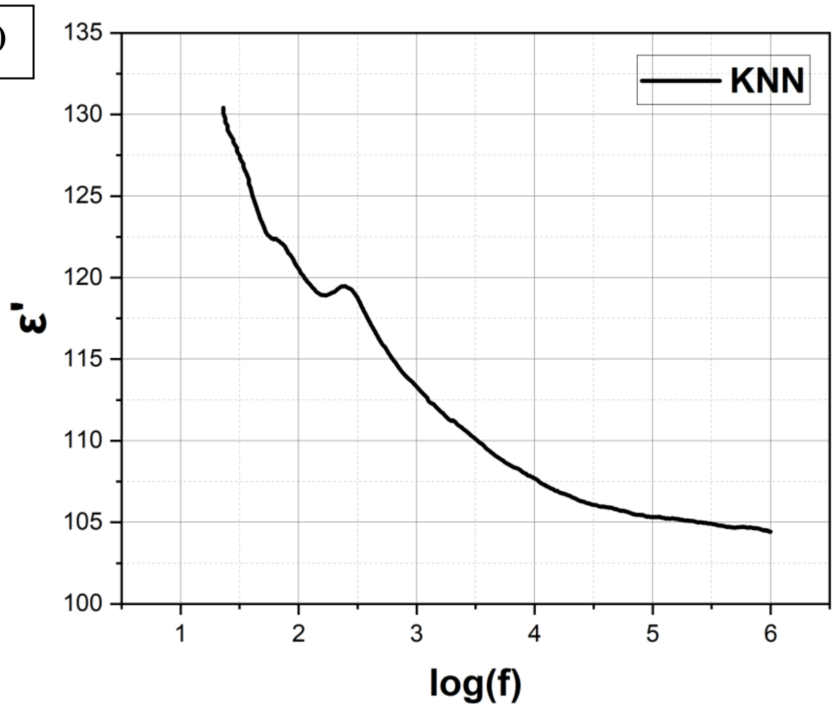
5.2.1 Frequency-Dependent Dielectric Properties

The frequency-dependent dielectric behavior of KNN, BCHfT, KNN–1 wt% BCHfT and KNN–1.5 wt% BCHfT ceramics was investigated in order to understand the polarization response, dielectric loss and charge transport behavior of the prepared lead-free ceramic system.

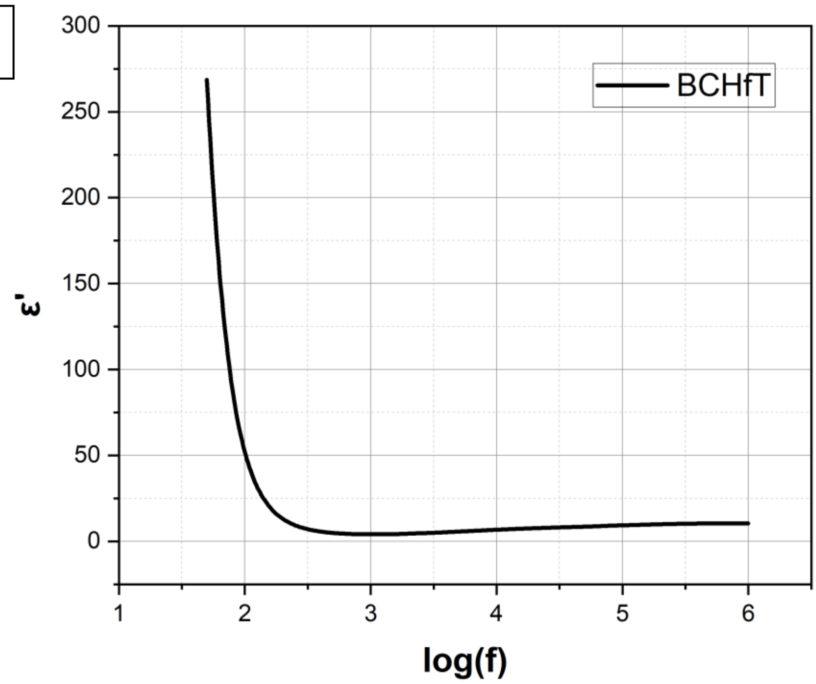
The real part of permittivity, ε', imaginary part of permittivity, ε'', dielectric loss tangent, $\tan\delta$, AC conductivity, σ_{ac}, and impedance response were analyzed as a function of frequency.

For all compositions, ε' decreased with increasing frequency. **(** ***Figure 5.2.1*** **)**

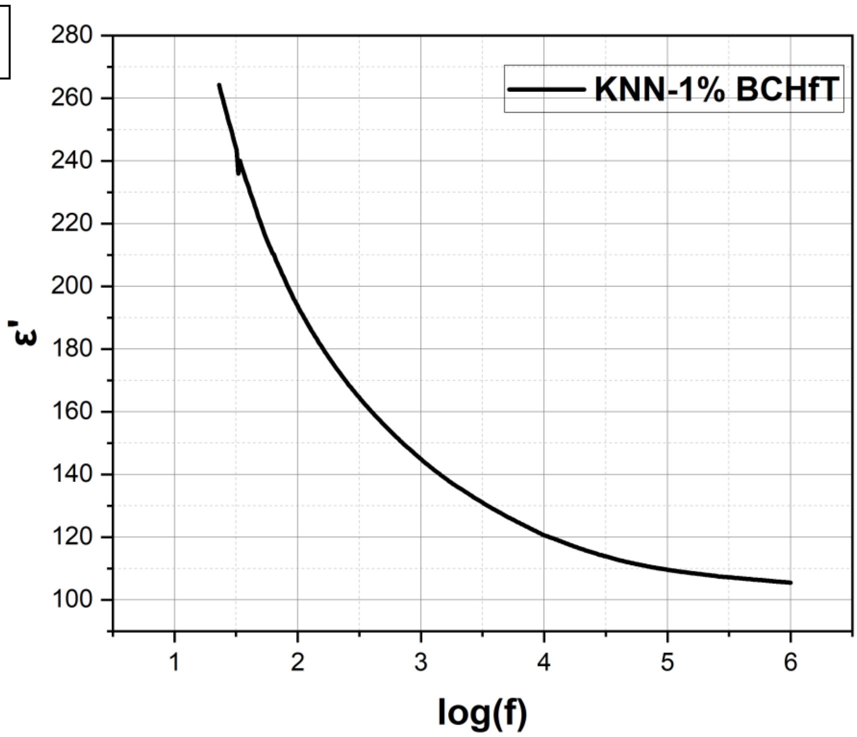
(a)


KNN
135
130
125
120
115
110
105
100
ε'
1
2
3
4
5
6
log(f)

(b)


BCHfT
300
250
200
150
100
50
0
ε'
1
2
3
4
5
6
log(f)

(c)



(d)

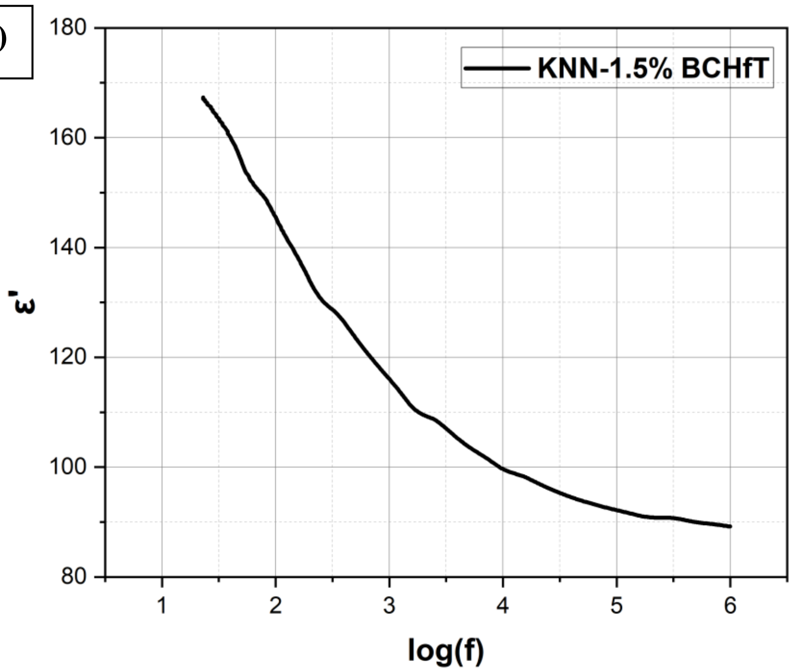


Figure 5.2.1: Dielectric constant (real part) vs logarithmic frequency a) pure KNN b) pure BCHfT c) KNN-1%BCHfT d) KNN-1.5%BCHfT

This behavior is typical for ferroelectric and dielectric ceramic materials and can be explained by the frequency dependence of polarization mechanisms.

At low frequency, electronic, ionic, dipolar and space-charge polarization can contribute to the total dielectric response. In particular, charge accumulation at grain boundaries, interfaces, defects and electrode regions can produce a high apparent dielectric constant. However, when the frequency increases, slower polarization mechanisms such as space-charge and interfacial polarization cannot follow the rapidly alternating electric field. As a result, the total polarization decreases and the value of ε′ becomes lower at high frequency.[44]

- The pure KNN sample showed comparatively low but stable permittivity. The ε′ value decreased only moderately with frequency, indicating a relatively stable dielectric response. The dielectric loss of KNN was also comparatively low, suggesting that pure KNN has lower leakage contribution than BCHfT.
- However, the relatively low ε′ value indicates limited polarization capability in the unmodified KNN ceramic. The BCHfT sample exhibited very high ε′ at low frequency, followed by a sharp decrease with increasing frequency. This indicates that BCHfT possesses strong low-frequency polarization, which may arise from interfacial polarization, grain-boundary effects and space-charge accumulation.

However, the very high dielectric loss and broad tanδ maximum indicate that the BCHfT ceramic is also significantly lossy. **(**Error! Reference source not found.**.2.2)**

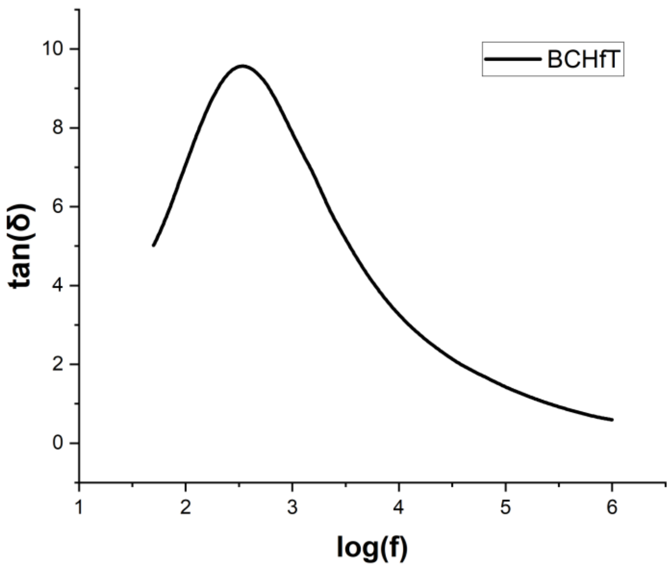


Figure 5.2.2: Dielectric loss vs logarithmic frequency of BCHfT

Therefore, although BCHfT has strong polarizability, its direct use as a dielectric/piezoelectric component may be limited by high loss and conduction-related effects.

The KNN–1 wt% BCHfT composite showed improved dielectric permittivity compared with pure KNN while maintaining a more acceptable dielectric loss than pure BCHfT. The increase in ε′ suggests that the addition of a small amount of BCHfT enhanced the polarizability of the KNN matrix. (***Figure 5.2.3***)

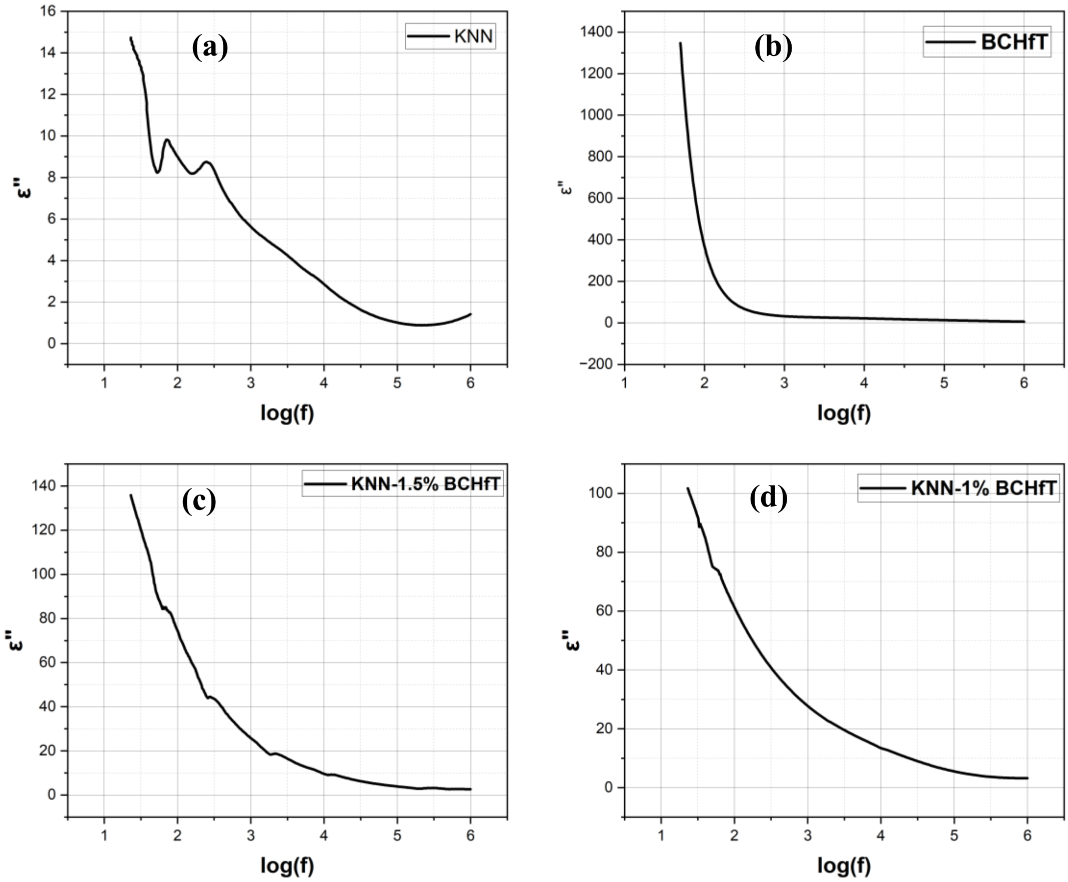


Figure 5.2.3: epsilon imaginary (absorption) vs logarithmic frequency. a) pure KNN b) pure BCHfT c) KNN-1.5%BCHfT d) KNN-1%BCHfT

This improvement may be associated with local lattice distortion, interfacial polarization between the KNN and BCHfT phases, improved domain-wall response and possible modification of defect structure.

The dielectric loss decreased continuously with frequency, indicating that the loss contribution becomes less significant at high frequency. **(Figure 5.2.4)**

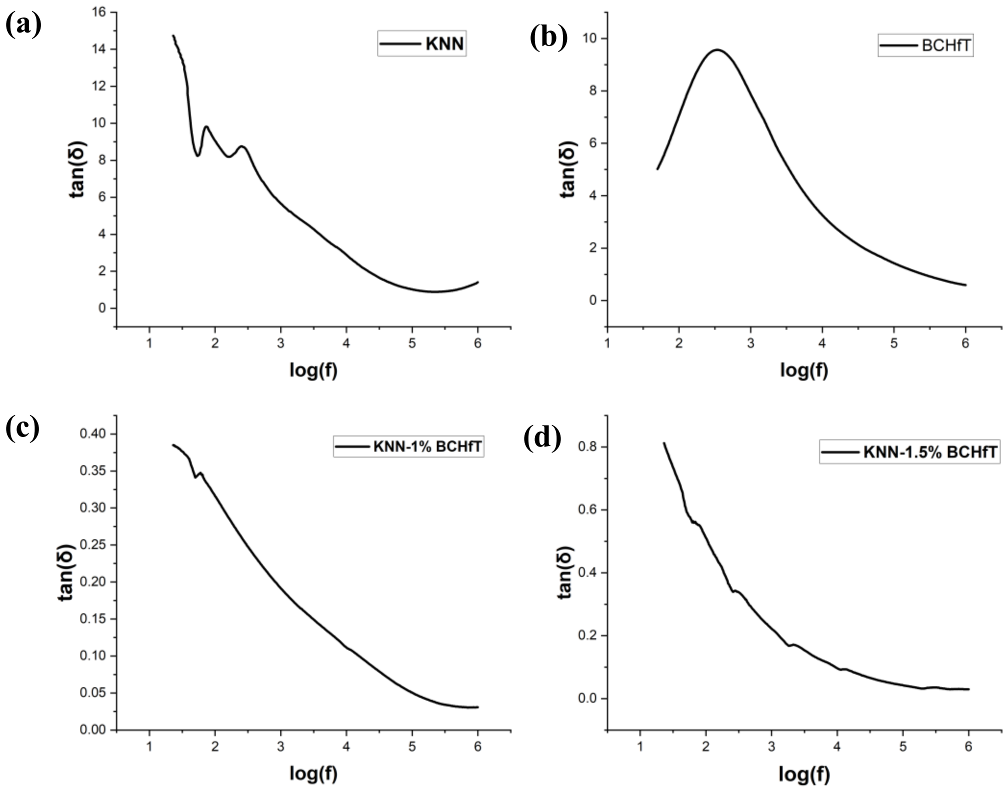


Figure 5.2.4: Dielectric loss of a)KNN b)BCHfT c) KNN-1%BCHfT d) KNN-1.5%BCHfT

Among the investigated compositions,

1) KNN–1% BCHfT showed the best balance between enhanced dielectric response and moderate dielectric loss.
2) For KNN–1.5% BCHfT, ε′ also decreased with frequency, but the low-frequency tanδ was higher than that of KNN–1 wt% BCHfT.

This suggests that increasing the BCHfT content from 1 wt% to 1.5 wt% may introduce stronger interfacial polarization and defect-related conduction.

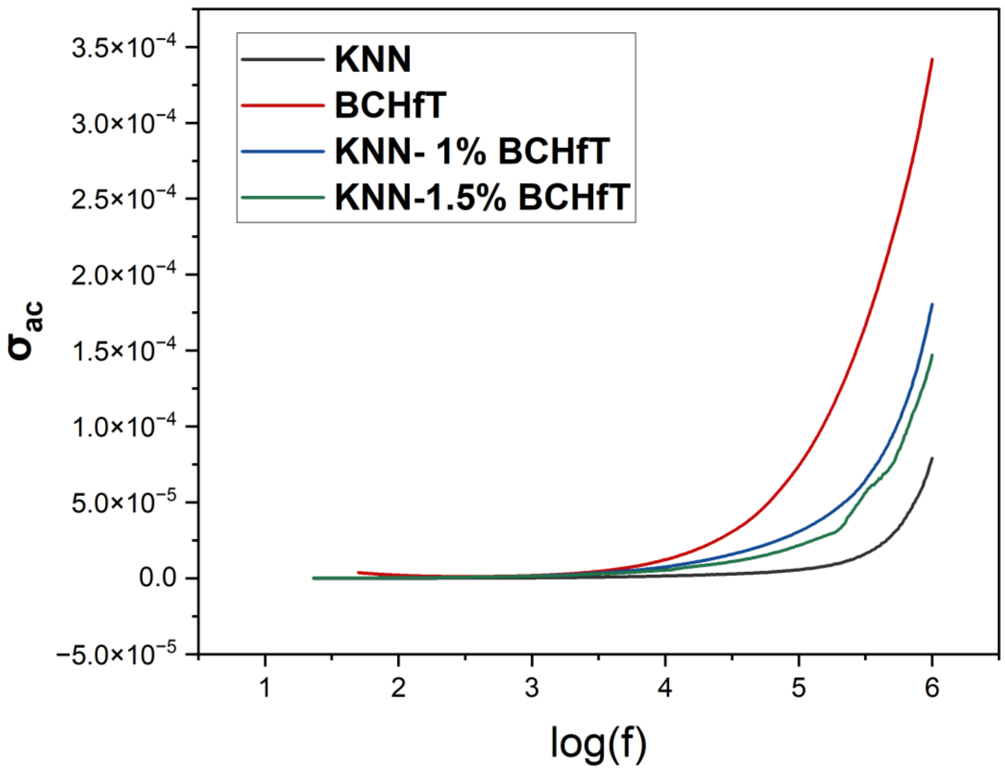


Figure 5.2.5: Comparison sigma curve for pure KNN,BCHfT, KNN-1%BCHfT and KNN-1.5%BCHfT

Although the composite still shows dielectric dispersion, the higher initial loss indicates that excessive BCHfT addition may increase leakage or grain boundary contribution. Therefore, 1 wt% BCHfT appears to be a more suitable addition level than 1.5 wt% for achieving a balanced dielectric response.

The AC conductivity increased with increasing frequency for all compositions.(***Figure 5.2.5***)

This trend indicates frequency-assisted charge transport and hopping conduction. At higher frequency, charge carriers can respond through localized hopping, resulting in increased σ_{ac}. The conductivity rise was most pronounced in samples with stronger interfacial or defect-related polarization. This confirms that dielectric loss and conductivity are closely connected in the present ceramic system.

Some irregular points were observed in the frequency-dependent ε″, tanδ and conductivity curves. These may originate from instrument range switching, contact instability, electrode polarization, data-conversion error or local defect-related relaxation. Therefore, sharp isolated dips or spikes should not be interpreted as major phase transitions unless they are reproducible in repeated measurements.

In the final analysis, the continuous trend of the curves should be emphasized rather than isolated anomalous data points. Overall, the frequency-dependent dielectric results confirm that BCHfT addition modifies the dielectric response of KNN. The 1 wt% BCHfT-added sample shows enhanced permittivity compared with pure KNN and lower loss than BCHfT-rich behavior. Therefore, KNN–1 wt% BCHfT can be considered the most promising composition among the tested samples from the viewpoint of dielectric balance.

5.2.1.1 Complex Impedance and Nyquist Plot Analysis

Complex impedance spectroscopy was used to study the electrical response of KNN, BCHfT, KNN–1 wt% BCHfT and KNN–1.5 wt% BCHfT ceramics. The Nyquist plot was drawn by plotting the real part of impedance, Z', on the X-axis and the negative imaginary part of impedance, $-Z''$, on the Y-axis. This representation is useful for understanding the contribution of resistive and capacitive components in electroceramic materials. In polycrystalline ceramics, the impedance response may arise from

1) grain interior,
2) grain boundary,
3) electrode interface and
4) defect-related polarization processes.

The Nyquist plots of the investigated samples do not show perfect ideal semicircles. Instead, the plots show depressed, distorted and incomplete arcs. (**Error! Reference source not found.**) This indicates that the electrical relaxation process is non-Debye type in nature. In an ideal Debye relaxation process, a single relaxation time gives a perfect semicircle. However, real ceramic materials contain grains of different size, grain boundaries, pores, defects, oxygen vacancies and electrode interfaces. These microstructural and defect-related factors produce a distribution of relaxation times, resulting in non-ideal Nyquist behavior.[45]

The pure KNN sample shows a large open arc with very high impedance. The imaginary component of impedance is significantly larger than the real component, indicating a strongly capacitive response. The arc is not completed within the measured frequency range, which suggests that the relaxation process is not fully covered by the experimental frequency window. The high impedance of KNN indicates good insulating behavior, but the distorted nature of the curve suggests overlapping relaxation contributions from grain boundaries and defect-related polarization.

The BCHfT sample shows a much lower impedance value compared with KNN and the KNN–BCHfT composites. This indicates that BCHfT is more conductive or more lossy under the present measurement condition. The Nyquist curve of BCHfT shows a depressed arc along with a low-frequency tail. The depressed arc indicates non-Debye relaxation, while the low-frequency tail may be associated with electrode polarization, space-charge accumulation or interfacial charge movement. This result agrees with the high dielectric loss and strong low-frequency dielectric response observed in the frequency-dependent dielectric plots of BCHfT.

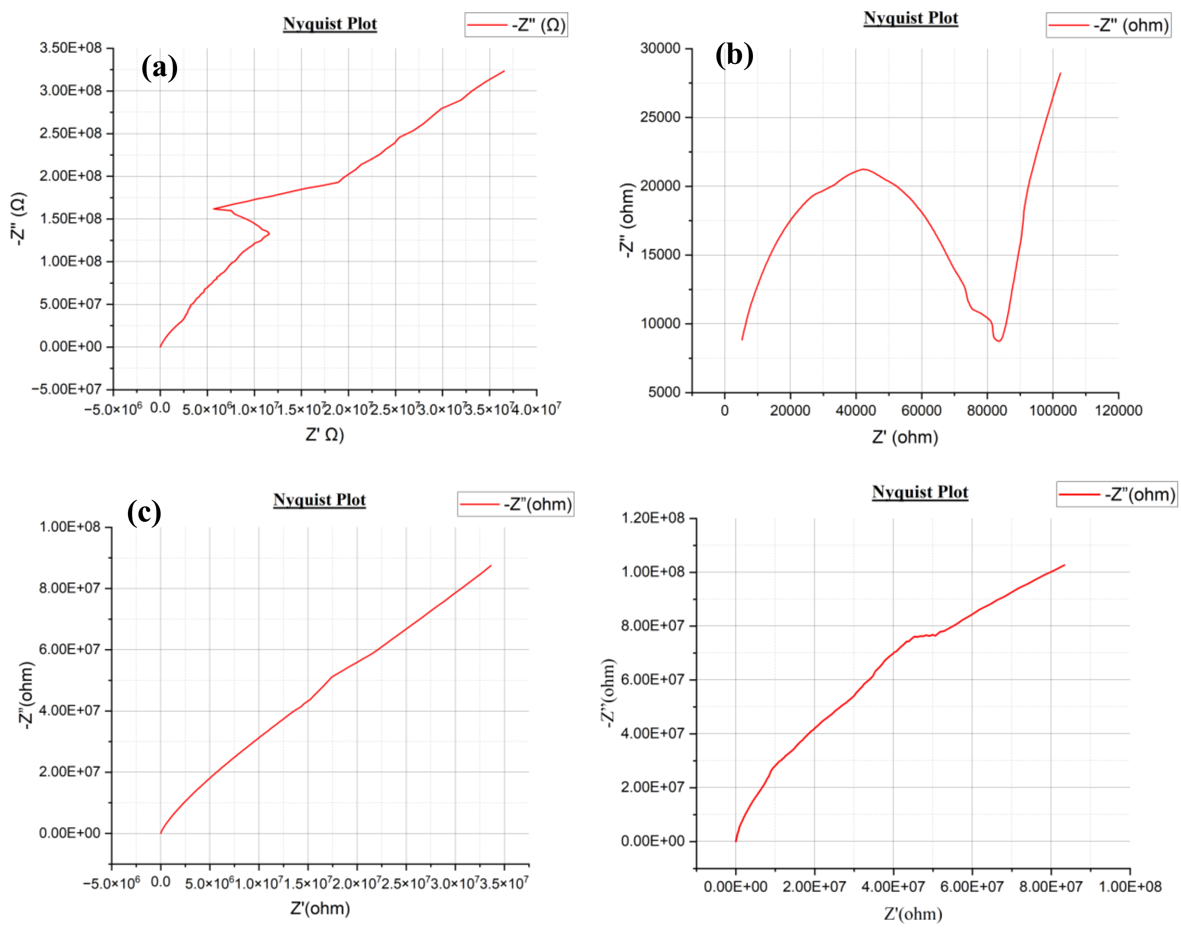


Figure 5.2. 6: Nyquist plot. a)KNN b)BCHfT c)KNN-1%BCHfT d)KNN-1.5%BCHfT

For the KNN–1% BCHfT composite, the Nyquist plot shows an open capacitive arc. Compared with pure KNN, the maximum value of $-Z''$ is reduced, indicating modification of the dielectric relaxation behavior due to BCHfT addition. The addition of a small amount of BCHfT may introduce local lattice distortion, interfacial polarization and improved polarizability in the KNN matrix. At the same time, the loss remains more controlled than in pure BCHfT. Therefore, KNN–1 wt% BCHfT shows a better balance between dielectric response and impedance behavior.

The KNN–1.5 wt% BCHfT sample shows a higher impedance scale than the KNN–1 wt% BCHfT sample, but its Nyquist plot is more non-ideal. A kink or plateau-like region is observed in the curve, indicating possible overlapping relaxation processes. This may be due to increased interfacial polarization, defect accumulation or grain-boundary heterogeneity caused by the higher BCHfT content. Although the impedance is high, the dielectric loss at low frequency is also higher than that of KNN–1 wt% BCHfT. Therefore, excessive BCHfT addition may increase non-ideal relaxation and defect/interface-related loss.

The absence of well-resolved complete semicircles in the Nyquist plots means that exact grain and grain-boundary resistance values cannot be extracted directly without equivalent-circuit fitting. A suitable equivalent circuit for such non-ideal ceramic systems may include grain and grain-boundary resistance elements connected with constant phase elements. The use of constant phase elements is more appropriate than ideal capacitors because the observed Nyquist plots show non-ideal capacitive behavior.

Overall, the Nyquist plot analysis confirms that BCHfT addition significantly modifies the impedance response of KNN. Pure KNN behaves as a highly resistive and strongly capacitive ceramic, while BCHfT shows comparatively low impedance and high loss contribution. The KNN–1 wt% BCHfT composite provides the most balanced impedance response among the investigated compositions. The 1.5 wt% BCHfT addition increases non-ideal relaxation behavior and may introduce stronger interfacial or defect-related effects. Therefore, from the combined dielectric and impedance analysis, KNN–1 wt% BCHfT can be considered the most suitable composition among the tested samples for achieving improved dielectric response with controlled loss behavior.

So, the findings are:

1. BCHfT addition significantly modifies the impedance response of KNN. Pure KNN behaves as a highly resistive and strongly capacitive ceramic, while BCHfT shows comparatively low impedance and high loss contribution.
2. The KNN–1 wt% BCHfT composite provides the most balanced impedance response among the investigated compositions.
3. The 1.5 wt% BCHfT addition increases non-ideal relaxation behavior and may introduce stronger interfacial or defect-related effects.

Therefore, from the combined dielectric and impedance analysis, KNN–1 wt% BCHfT can be considered the most suitable composition among the tested samples for achieving improved dielectric response with controlled loss behavior.

5.2.2 Temperature-Dependent Dielectric Properties of KNN–1 wt% BCHfT

The temperature-dependent dielectric behavior of KNN–1 wt% BCHfT was studied at different frequencies in order to identify dielectric stability, thermally activated polarization, phase-transition-related anomalies and high-temperature loss behavior.

The real permittivity ε' remained relatively low and stable at lower temperature, especially at higher measurement frequencies. (**Error! Reference source not found.**) This indicates that the composite ceramic maintains a comparatively stable dielectric response before the onset of strong thermally activated processes. With increasing temperature, a strong frequency-dependent rise in ε' was observed, especially at low frequency. The increase became significant above approximately 220–250°C. The 100 Hz curve showed the highest permittivity, while the 1 MHz curve remained comparatively low and stable. The maximum ε' values obtained were 714.10 at 540°C for 100 Hz, 298.77 at 520°C for 1 kHz, 85.76 at 520°C for 10 kHz, and 30.63

at 520°C for 1 MHz. This strong frequency dispersion indicates that the high-temperature dielectric response is strongly influenced by space-charge polarization, grain-boundary effects and thermally activated charge movement. The response is therefore not purely intrinsic ferroelectric polarization.

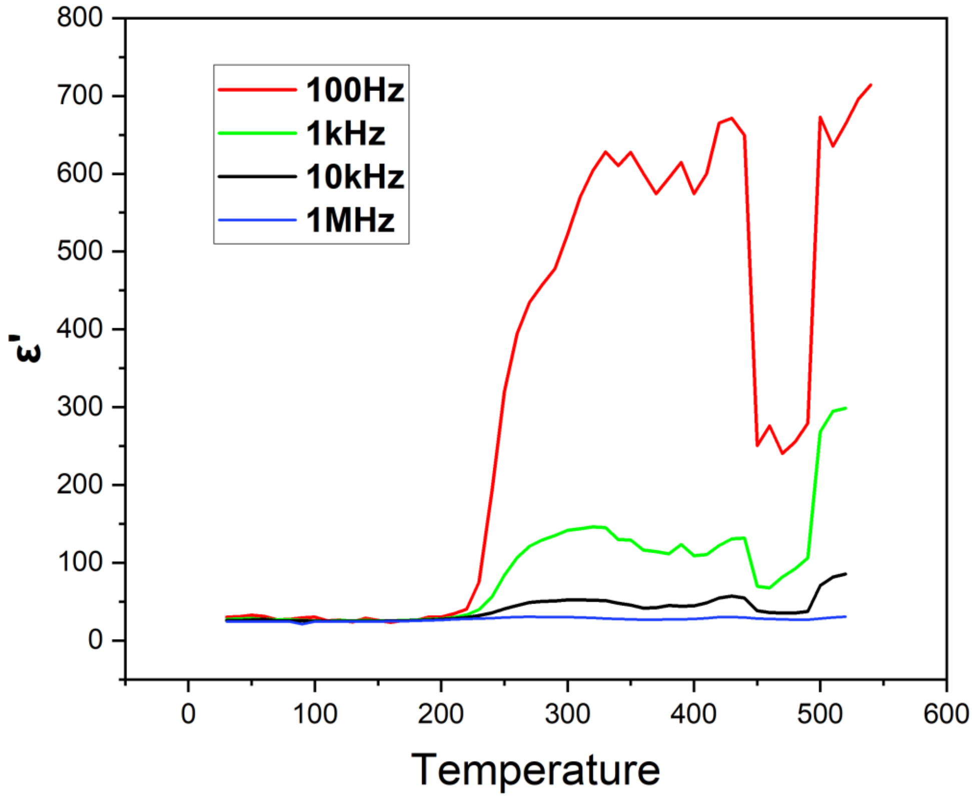
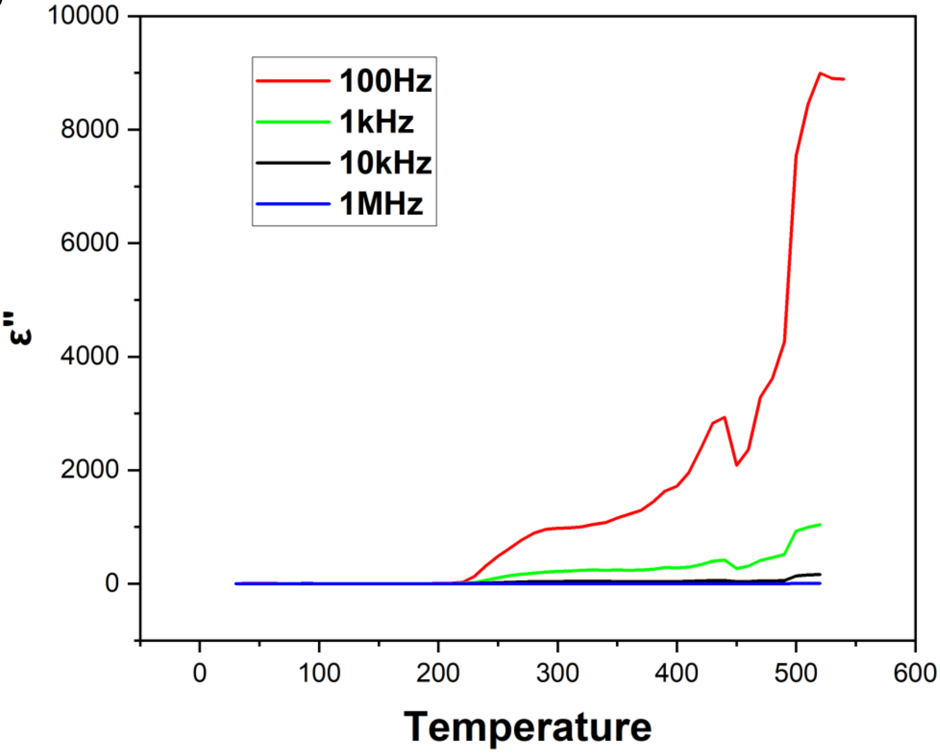


Figure 5.2.7: Real permittivity ε′ at varying frequency with increasing temperature.

The imaginary permittivity ε″ and dielectric loss tangent tanδ also increased strongly at elevated temperature. (**Error! Reference source not found.**) The increase was very large at low frequency, particularly above about 400–450°C. The maximum tanδ values were 15.255 at 490°C for 100 Hz, 5.061 at 470°C for 1 kHz, 1.966 at 500°C for 10 kHz, and 0.2279 at 520°C for 1 MHz. This behavior indicates significant dielectric loss and leakage-related contribution at high temperature. The sharp rise in ε″ and tanδ may be caused by increased mobility of oxygen vacancies, alkali-related vacancies, trapped charges and grain-boundary charge carriers. Therefore, the high-temperature dielectric anomaly should be described as a diffuse dielectric anomaly coupled with thermally activated conduction rather than a clean and sharp ferroelectric Curie transition.

(a)



(b)

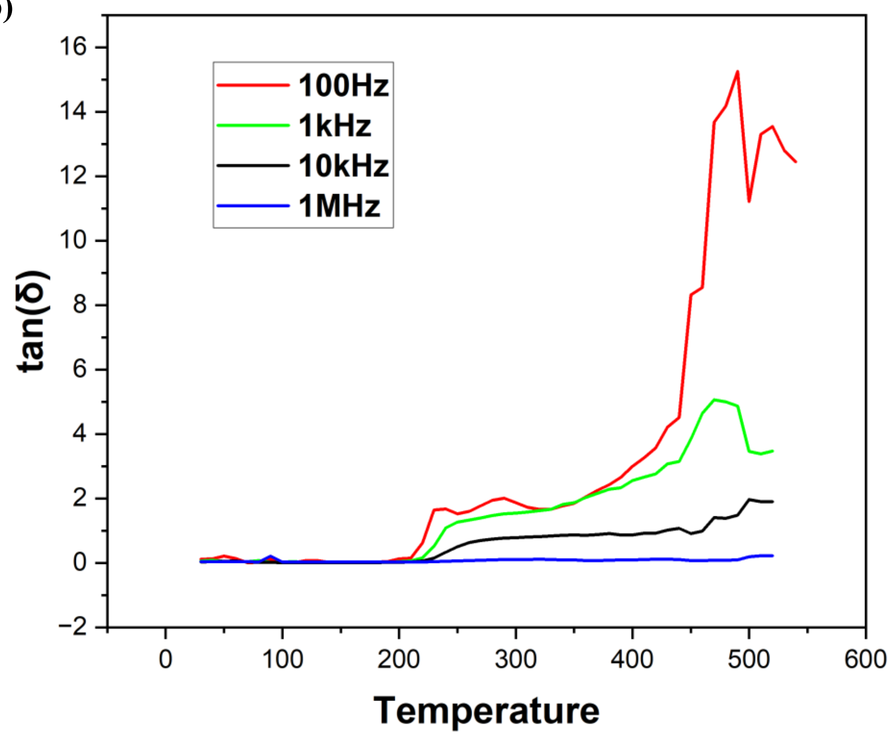


Figure 5.2.8: Dielectric coefficient (imaginary) and loss tangent vs temperature

The impedance and resistance results further support this interpretation.

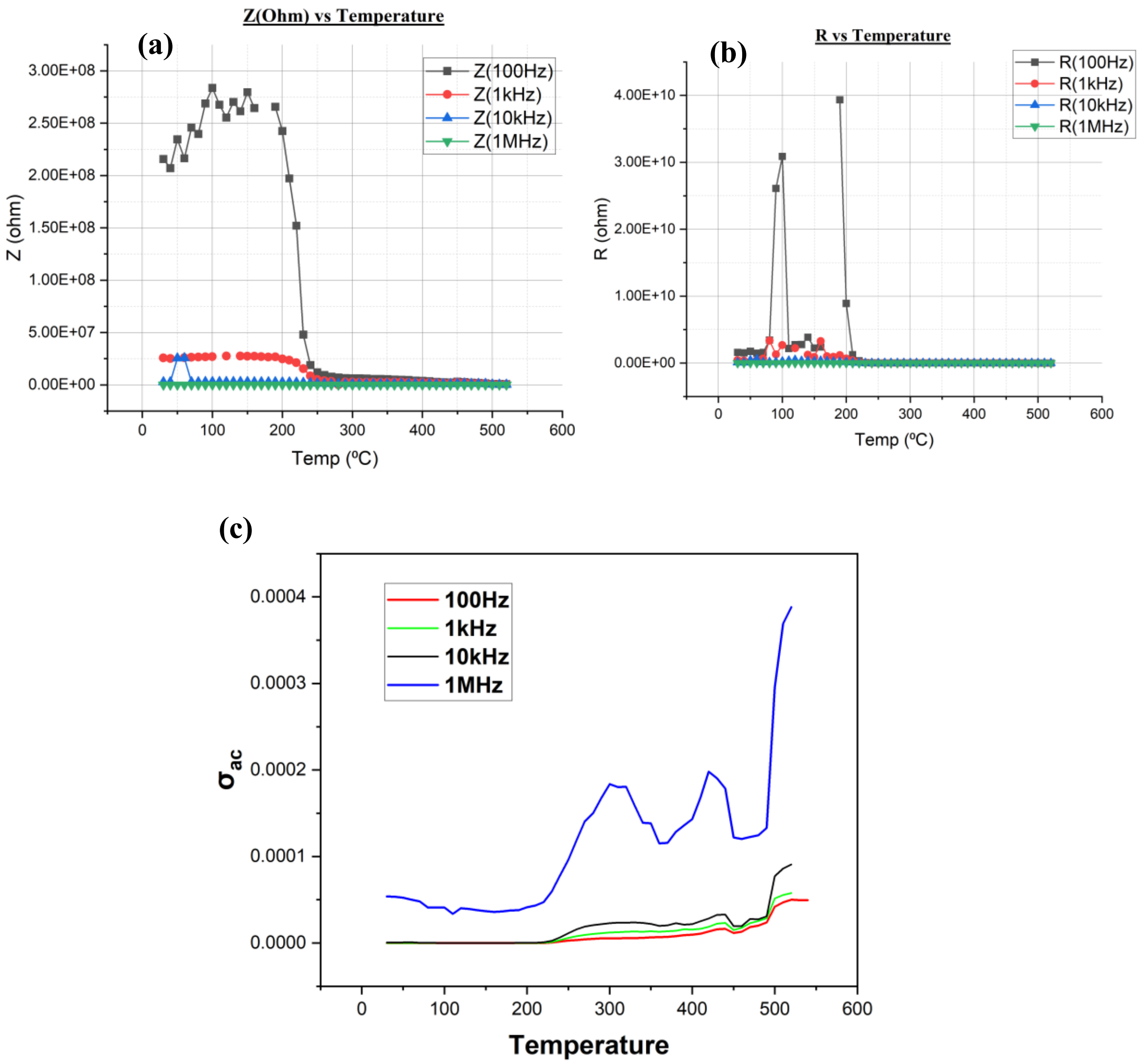


Figure 5.2.9: a) Impedence vs Temperature b) Resistance vs Temperature and c) Conductivity vs Temperature

At lower temperature, the impedance was high, indicating insulating behavior. With increasing temperature, the impedance decreased sharply, especially above about 200–250°C. This decrease confirms the increase in electrical conductivity at elevated temperature. The AC conductivity also increased with temperature and frequency, showing thermally activated conduction behavior. Such behavior is common in oxide ferroelectric ceramics where defect migration and hopping conduction become stronger at high temperature.

Several anomalies were observed in the temperature-dependent data. The first anomaly appears near 220 250°C, where ε′, ε″ and tanδ begin to increase rapidly. This may be associated with KNN-related polymorphic transition behavior and enhanced interfacial polarization. In addition, abnormal spikes in resistance and AC conductivity at selected temperatures are likely due to experimental instability, electrode/contact effects, instrument range switching or data-processing artifacts. These points should be discussed carefully and should not be overinterpreted as intrinsic phase transitions.

5.2.2.1 Defect Mechanisms and Arrhenius Analysis

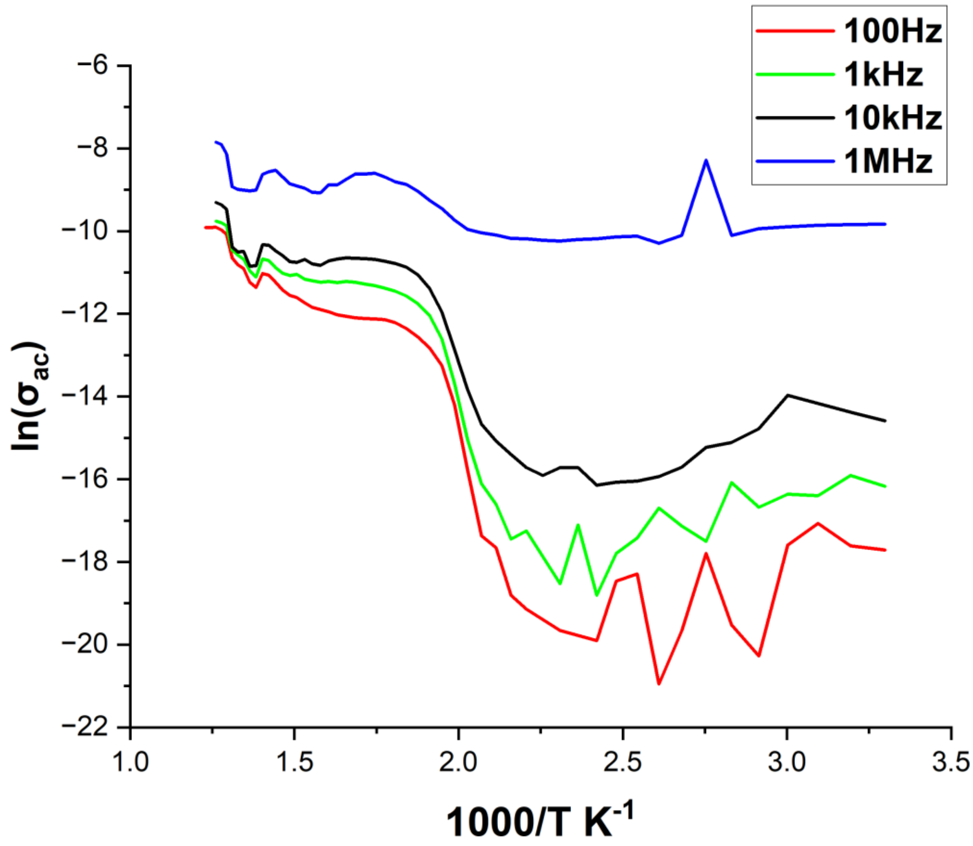


Figure 5.2.10: Arrhenius plot for defect mechanism analysis in KNN-1% BCHfT

An Arrhenius plot is generated using data taken from dielectric studies carried out as a function of temperature. In particular, by plotting the natural logarithm of a dielectric constant (e.g., ac conductivity or relaxation frequency) versus the reciprocal temperature (1000/T), one will generate a plot that demonstrates a clear trend of linearity. The existence of such linearity reveals that the dielectric loss at high temperatures is indeed thermally activated in accordance with the Arrhenius empirical relationship.

One of the important parameters that can be derived using such a method is the activation energy, Ea, which will be determined directly from the value of the slope. The slope value, itself, represents the energy barrier associated with migration through the perovskite lattice.

To calculate activation energy from Arrhenius plot, slope is measured using origin pro software and calculated below:

Frequency	Slope	$E_a = -slope \times k_B \times 1000$
100 Hz	-33.71	2.91 eV
1kHz	-34.64	3 eV
10kHz	-13.03	1.12 eV
1MKz	-4.511	0.4 eV

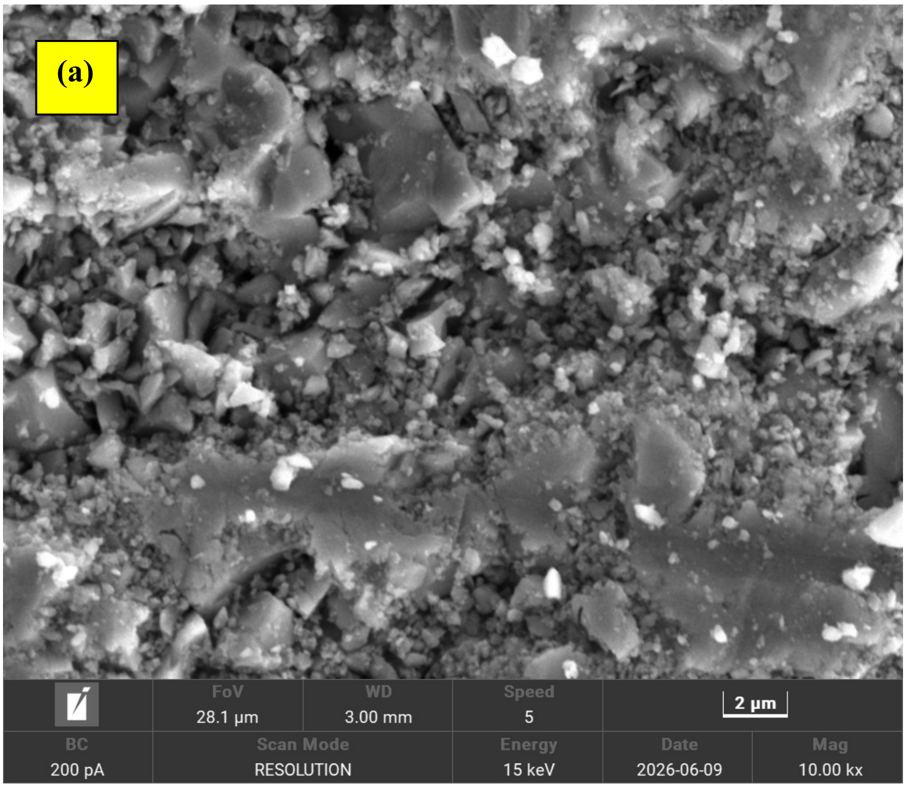
Table 5.2.1: Activation energy calculation from Arrhenius plot.

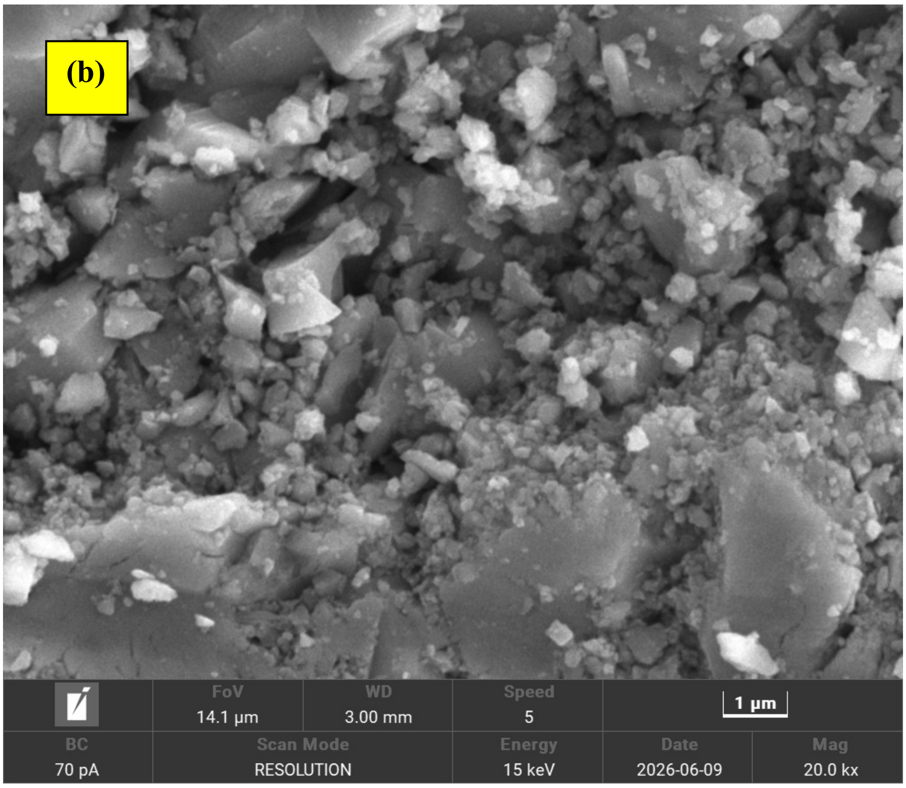
Based on the linear fit of the curve in the high-temperature regime, the obtained activation energies were 2.90 eV, 2.99 eV, 1.12 eV, and 0.39 eV for 100 Hz, 1 kHz, 10 kHz, and 1 MHz, respectively.. This specific energy range is a well-established signature of oxygen vacancy migration. It indicates that the long-range hopping of doubly ionized oxygen vacancies ($V_O^{\bullet\bullet}$)-which are intrinsically generated to compensate for alkali volatilization during the high-temperature sintering process is the primary mechanism driving the dielectric relaxation and conductivity at elevated temperatures. Consequently, the curve demonstrates that the material's high-temperature dielectric stability is fundamentally governed by the concentration and mobility of these oxygen vacancies.

5.2.3 Overall Finding

The combined frequency- and temperature-dependent dielectric investigations show that BCHfT addition significantly modifies the dielectric behavior of KNN. The addition of 1 wt% BCHfT enhances the dielectric permittivity of KNN while keeping the dielectric loss within a more acceptable range than higher BCHfT-rich response. The 1.5 wt% BCHfT composition shows higher low-frequency loss, suggesting that excessive BCHfT addition may increase defect-related or interfacial polarization loss. The temperature-dependent test of KNN–1 wt% BCHfT reveals a diffuse dielectric anomaly and strong high-temperature loss behavior, indicating the presence of thermally activated charge transport. Therefore, KNN–1 wt% BCHfT is the most promising composition among the tested samples, but further optimization of sintering, density, electrode quality and defect control is required to improve high-temperature dielectric stability.

5.3 SEM Analysis





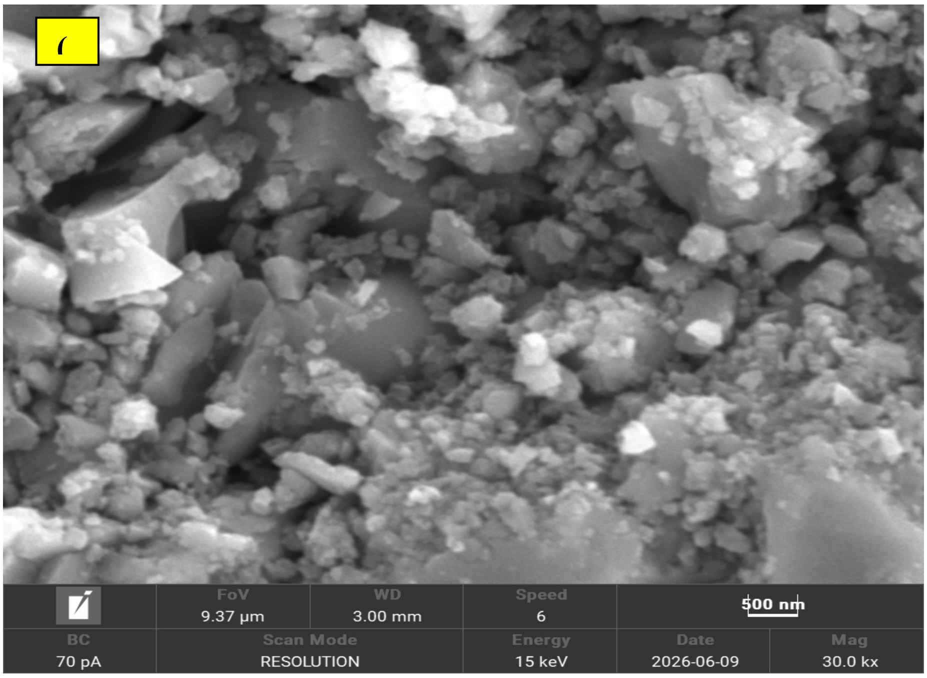


Figure 5.3.1: SEM image of KNN-1.5% BCHfT at a)10000x b)20000x and c)30000x

SEM images reveal that the samples have an agglomerated and inhomogeneous microstructure with inhomogeneity in the sizes of the particles or grains and interstitial porosity between the particles. The presence of fine particles in between large angular particles suggests that the sample is not fully dense. These features of the microstructure might lead to enhanced polarizations due to defects and grain boundaries, which might be responsible for dielectric loss. The above microstructure resembles the structure of KNN-1.5 wt% BCHfT, where an excess amount of BCHfT may cause enhanced grain boundary polarization compared to KNN-1 wt% BCHfT.

5.4 Piezoelectric Co-efficient(d_{33}) Analysis

Composition	Unpoled d_{33} value
KNN	13
KNN-1%BCHfT	15.5
KNN-1.5%BCHfT	12

Table 5.4.1: The unpoled d33 data for synthesized ceramics

As the samples were not poled, the dipoles didn't align in the same direction and so the d33 values are random and can't be used with certainty. For every sample, at 4 points d33 were measured and the average d33 value is shown.

Though KNN-1%BCHfT composition showed increased d33 compared to pure KNN samples without poling, it is not reliable and certain that poled sample will also follow this trend. But analyzing all the data previously we can expect the trend will continue in the poled sample also.

After completing poling, at voltage up to 5kV/mm at 120^{0}C for 20mins. After poling, the samples were aged for 24 hours before measurement and the dipoles will be aligned in the same direction and d33 data will be taken again. The values will increase and then the data can be used with certainty.

Chapter:6 - Conclusion

In this thesis, lead-free KNN–BCHfT piezoelectric ceramics were successfully prepared and investigated with the aim of improving the dielectric and piezoelectric behavior of KNN-based ceramics through the incorporation of BCHfT. The study was motivated by the need to develop environmentally friendly alternatives to lead-based PZT ceramics, which are widely used in sensors, actuators, transducers, and energy-harvesting devices but raise serious environmental and health concerns due to their lead content. KNN was selected as the main lead-free piezoelectric matrix because of its perovskite structure, ferroelectric nature, and potential piezoelectric response. However, pure KNN suffers from several processing and performance limitations, such as alkali volatilization, poor densification, defect formation, and unstable piezoelectric properties. Therefore, BCHfT was introduced as a $BaTiO_3$-based lead-free ferroelectric modifier to improve the structural, dielectric, and piezoelectric behavior of KNN ceramics.

Pure KNN, pure BCHfT, KNN–1 wt% BCHfT, and KNN–1.5 wt% BCHfT ceramics were prepared by the conventional solid-state reaction method. The prepared samples were characterized by X-ray diffraction, frequency and temperature-dependent dielectric measurements, conductivity analysis, impedance-related analysis, and d_{33} measurement. The XRD analysis confirmed the formation of crystalline phases in the synthesized ceramics. Pure KNN showed the presence of potassium sodium niobate as the major phase along with a K-Nb-O related secondary phase. Pure BCHfT showed $BaTiO_3$ related, $CaTiO_3$-related, and Hf-Ti-O related phases. The composite samples indicated that BCHfT addition influenced the phase constitution of the KNN-based system, suggesting structural modification due to composite formation. These observations support the idea that BCHfT addition can affect the crystal structure and phase development of KNN based lead-free ceramics.

The dielectric results clearly showed that the dielectric response of all samples was strongly dependent on frequency. The real part of dielectric permittivity decreased with increasing frequency, which can be explained by the reduced contribution of slow polarization mechanisms such as space-charge and interfacial polarization at higher frequencies. At lower frequencies, charge carriers and interfacial polarization can follow the applied alternating field, resulting in higher dielectric permittivity. However, at higher frequencies, these slower polarization processes cannot respond effectively, causing a decrease in dielectric constant. This behavior confirms the presence of frequency-dependent polarization mechanisms in the prepared ceramics.

Among the investigated compositions, KNN-1% BCHfT showed the most balanced dielectric performance. The addition of 1 wt% BCHfT improved the dielectric permittivity of KNN while maintaining dielectric loss within a comparatively acceptable range. This indicates that a small amount of BCHfT can positively modify the dielectric behavior of KNN, possibly by influencing phase structure, polarization response, and microstructural characteristics. In contrast, the KNN-1.5% BCHfT composition showed higher dielectric loss, especially at low frequency. This higher loss suggests that excessive BCHfT addition may increase defect-related polarization, interfacial polarization, leakage contribution, or grain-boundary effects. Therefore, the results indicate that BCHfT addition is beneficial only up to an optimum level, and excessive addition may deteriorate the dielectric performance.

The temperature-dependent dielectric analysis of KNN-1% BCHfT revealed diffuse dielectric behavior and thermally activated charge transport at elevated temperatures. The increase in dielectric loss and conductivity at higher temperature suggests that mobile charge carriers, particularly oxygen vacancies and other defect-related species, contribute to dielectric relaxation and conduction. This behavior is commonly associated with defect migration, space-charge polarization, and thermally activated hopping conduction in perovskite ceramics. Therefore, although KNN-1% BCHfT showed improved dielectric behavior compared with the other investigated compositions, further improvement in high-temperature dielectric stability is still necessary.

The piezoelectric coefficient measurement also supported the beneficial effect of 1 wt% BCHfT addition. The measured unpoled d_{33} values were 13 pC/N for pure KNN, 15.5 pC/N for KNN-1% BCHfT, and 12 pC/N for KNN-1.5% BCHfT. Although these values were obtained from unpoled samples and therefore cannot represent the final optimized piezoelectric performance, the result still indicates that KNN-1% BCHfT has a comparatively better piezoelectric response than pure KNN and KNN-1.5% BCHfT. Since the samples were not poled, the dipoles were not fully aligned, and the measured d_{33} values should be considered preliminary. Proper poling treatment is expected to align the ferroelectric domains and provide more reliable piezoelectric data.

Overall, this study demonstrates that BCHfT addition can modify the dielectric and piezoelectric behavior of KNN based lead-free ceramics. Among the studied compositions, KNN-1% BCHfT is the most promising composition because it shows improved dielectric permittivity, comparatively acceptable dielectric loss, and the highest unpoled d_{33} value. On the other hand, increasing BCHfT content to 1.5 wt% appears to increase dielectric loss and reduce the piezoelectric response, indicating that excessive BCHfT addition may introduce unfavorable defect related or interfacial effects.

Finally, the present work confirms that KNN-BCHfT composite formation is a useful approach for developing environmentally friendly lead-free piezoelectric ceramics. However, further optimization is required to achieve better functional performance. Future work should focus on improving sintering conditions, controlling density and porosity, reducing alkali volatilization and oxygen vacancy related defects, improving electrode quality, and applying proper poling treatment. With these improvements, KNN-1% BCHfT ceramics may become more suitable for future lead-free piezoelectric applications.

Key Aspects Addressed in This Study

The present research on KNN-BCHfT lead-free piezoelectric ceramics addressed several important aspects related to synthesis, structural development, dielectric behavior, piezoelectric response, and future application potential. The major outcomes of this study can be summarized as follows:

Synthesis of Lead-Free KNN–BCHfT Ceramics:

Pure KNN, pure BCHfT, KNN-1% BCHfT, and KNN-1.5% BCHfT ceramics were successfully prepared using the conventional solid-state reaction method. The synthesis route was selected because it is simple, cost-effective, and widely used for ceramic processing. The prepared pellets were sintered at high temperature to develop the desired ceramic phases and improve the compactness of the samples.

Structural Phase Formation:

X-ray diffraction analysis confirmed the crystalline nature of the prepared ceramics. Pure KNN showed the formation of potassium sodium niobate as the major phase, while pure BCHfT showed $BaTiO_3$ based phase development with Ca- and Hf-related modifications. The KNN-BCHfT composite samples showed that BCHfT addition influenced the phase constitution of the KNN matrix, indicating that composite formation can modify the structural behavior of KNN-based ceramics.

Effect of BCHfT Addition:

The addition of BCHfT played an important role in modifying the dielectric and piezoelectric properties of KNN ceramics. Among the studied compositions, KNN-1% BCHfT showed the most balanced performance. This composition exhibited improved dielectric permittivity with comparatively acceptable dielectric loss, suggesting that a small amount of BCHfT can positively influence the polarization response of KNN ceramics.

Frequency-Dependent Dielectric Behavior:

The dielectric constant decreased with increasing frequency for all compositions. This behavior indicates that slow polarization mechanisms, such as space-charge polarization and interfacial polarization, contribute strongly at low frequency but cannot follow the rapidly alternating electric field at higher frequency. Therefore, the frequency-dependent dielectric response confirmed the presence of different polarization mechanisms in the prepared ceramics.

Dielectric Loss and Defect Contribution:

The dielectric loss behavior showed that KNN-1% BCHfT maintained comparatively better dielectric stability than KNN-1.5% BCHfT. The higher loss observed in the KNN-1.5% BCHfT sample may be related to stronger defect-related polarization, oxygen vacancies, interfacial charge accumulation, or conducting grain-boundary effects. This result indicates that excessive BCHfT addition may negatively affect dielectric performance.

Temperature-Dependent Dielectric Response:

Temperature-dependent dielectric analysis of KNN–1% BCHfT revealed diffuse dielectric behavior and thermally activated charge transport at elevated temperatures. The increase in conductivity and dielectric loss at higher temperature suggests that mobile charge carriers and defect species contribute to dielectric relaxation. This finding is important for understanding the high-temperature stability and defect mechanism of KNN-BCHfT ceramics.

Piezoelectric Response:

The measured unpoled d_{33} values were 13 pC/N for pure KNN, 15.5 pC/N for KNN–1% BCHfT, and 12 pC/N for KNN-1.5% BCHfT. The highest d_{33} value was obtained for KNN-1% BCHfT, indicating that this composition is the most promising among the studied samples. However, since the samples were unpoled, proper poling treatment is required to evaluate the final optimized piezoelectric performance.

Optimum Composition:

Based on the combined dielectric and piezoelectric results, KNN-1% BCHfT can be considered the optimum composition in this study. It showed improved dielectric permittivity, comparatively controlled dielectric loss, and the highest unpoled d_{33} value. On the other hand, increasing the BCHfT content to 1.5 wt% did not further improve the performance and instead increased dielectric loss.

Environmental and Application Significance:

This study contributes to the development of environmentally friendly lead-free piezoelectric ceramics as possible alternatives to lead-based PZT materials. The KNN-BCHfT system shows potential for future use in sensors, actuators, transducers, and energy-harvesting devices where stable dielectric and piezoelectric properties are required.

Prospects for Future Research:

The encouraging results of this study create opportunities for further investigation of KNN–BCHfT ceramics. Future work should focus on improved sintering conditions, better density control, reduction of alkali volatilization, defect minimization, improved electrode quality, microstructural analysis, and proper poling treatment. Further optimization may significantly enhance the piezoelectric response and make the KNN-BCHfT system more suitable for practical lead-free piezoelectric applications.

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